MILITARY SPECIFICATION

GENERATOR SET, DIESEL ENGINE ALTERNATING CURRENT

(NAVAL SUBMARINE USE)

1. SCOPE

1.1 This specification covers constant speed, alternating current, (a.c.) 60 hertz, diesel engine driven generator set for submarine auxiliary power under surface and snorkel operating conditions.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for the bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

F-F-351 Filters and Filter Elements Lubricating Oil, Internal Combustion Engines
O-A-548 - Antifreeze, Ethylene Glycol, Internal
O-1-490 - Inhibitor, Corrosion, Liquid Cooling System.
O-S-588 - Sodium Chromate, Anhydrous, Technical.
QQ-C-39 - Copper Alloy Castings (Including Cast Bar).
MMM-A-260 - Adhesive, Water-Resistant (for Sealing Waterproofed Paper).
MMM-A-1617 - Adhesive, Rubber Base, General Purpose.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.

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MIL-C-104 - Crates, Wood; Lumber and Plywood, Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-131 - Barrier Material, Water Vaporproof, Flexible, Heat-Sealable
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
MIL-C-915 - Cable and Cord Electrical, for Shipboard Use, General Specification For.
MIL-E-917 - Electric Power Equipment, Basic Requirements (Naval Shipboard Use.)
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-R-2729 - Regulator Sets, Voltage, A.C. Generator, Naval Shipboard Use.
MIL-G-3124 - Generator, Alternating Current, 60 Cycle (Naval Shipboard Use).
MIL-T-5624 - Turbine Fuel, Aviation,Grades JP-4 and JP-5
MIL-F-7194 - Filter, Reciprocating Engine, Inductionair, Aircraft.
MIL-L-9000 - Lubricating Oil, Shipboard, Internal Combustion Engine, High Output.
MIL-C-9959 - Container, Flexible, Reusable, Water-Vaporproof.
MIL-E-10062 - Engines, Preparation for Shipment and Storage of.
MIL-G-12803 - Gasket Material, Non-Metallic.
MIL-B-13239 - Barrier Material, Waterproofed, Flexible, All Temperatures.
MIL-M-15071 - Manuals, Technical Equipment and Systems, Content Requirements for.
MIL-C-15730 - Coolers, Fluid, Industrial, Naval Shipboard; Lubricating Oil, Hydraulic Oil and Fresh Water.
MIL-S-16032 - Switches, Shipboard Alarm Systems.
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-P-16304 - Pyrometers, Indicating.
MIL-B-16541 - Bronze, Valve: Castings.
MIL-M-16576 - Metal, Gun: Castings.
MIL-F-16884 - Fuel Oil, Diesel, Marine.
MIL-I-17244 - Indicators, Temperature, Direct-Reading, Bimetallic (3 and 5 Inch Dial).

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MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of.
MIL-I-18997 - Indicator, Pressure, Panel Mounted or Case Supported, General Specification.
MIL-S-19500 - Semiconductor Devices General Specification For.
MIL-T-19646 - Thermometers, Remote Reading, Self-Indicating Dial, Gas Actuated.
NIL-C-19836 - Coolers, Fluid, Industrial, Air Motor and Generator, Naval Shipboard.
MIL-F-20042 - Flanges, Pipe Bronze (Silver Brazing).
MIL-C-20159 - Copper-Nickel Alloy (70-30 and 90-19): Castings.
MIL-F-20627 - Filter Assembly and Filter Elements, Fluid Pressure (For Engines with Liquid Fuel Injection Systems).
MIL-F-20670 - Flanges, Pipe, Carbon Steel, 150 P.S.I., W.S.P. (For Naval Shipboard Use).
MIL-B-21230 - Bronze, Nickel Aluminum and Manganese - Nickel Aluminum: Castings, Ship Propeller Application.
MIL-L-21260 - Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-B-23921 - Bronze Nickel Aluminum Castings For Seawater Service.
MIL-P-24212 - Pressure Transducer Equipment (Electrical) (Naval Shipboard Use) .
MIL-W-24270 - Wells, For Indicators or Thermal Elements General Specification For.
MIL-D-24304 - Differential Pressure Transducer Equipment (Electrical) (Naval Shipboard Use).
MIL-M-24365 - Maintenance Engineering Analysis: Establishment of and Procedures and Formats for Associated Documentation; General Specification For.
MIL-T-24387 - Temperature Measurement Equipment Signal Conditioner and Power Supply (Electrical) (Naval Shipboard Use).
MIL-T-24388 - Thermocouples and Resistance Temperature Elements (Naval Shipboard Use).
MIL-I-24453 - Inhibitor, Corrosion, Soluble-Oil.
MIL-I-26860 - Indicator, Humidity, Plug, Color Change.
MIL-M-38761/2 - Microfilming and Photographing of Engineering/Technical Data and Related Documents: PCAM Card Preparation, Engineering Data Micro-Reproduction System: Microfilm Aperture and Tabulating Cards for Naval Ship Systems.
MIL-I-45208 - Inspection Systems Requirements.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-178 - Definitions Applicable to Speed-Governing of Electric Generator Sets.
MIL-STD-278 - Fabrication, Welding and Inspection; and Castings Inspection, and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-438 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Submarine Service.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipments) .
MIL-STD-471 - Maintainability Demonstration.
MIL-STD-701 - Lists of Standard Semiconductor Devices.
MIL-STD-721 - Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors and Safety.
MIL-STD-756 - Reliability Prediction.
MIL-STD-758 - Packaging Procedures for Submarine Repair Parts Utilizing Transparent Flexible Heat Sealable Film.
MIL-STD-785 - Requirements for Reliability Program (For Systems and Equipments) .
MIL-STD-882 - System Safety Program for Systems and Associated Sub-System and Equipment; Requirements for.

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MILITARY (Cont'd)
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Water-proofing; with Appropriate Test Methods.
MS16142 - Boss, Gasket, Seal Straight Thread Tube Testing, Standard Dimensions For.
MS35802 - Filter Elements, Fluid, Pressure-oil, Full-Flow.

HANDBOOK

MILITARY
MIL-HDBK-472 - Maintainability Prediction.

DRAWINGS

MILITARY
810-1385850 - Piping, Gage, for all service.
810-1385917 - Thermometer Selection Guide (General).
5000-S4501-64492 - Coolers, Lubricating Oil and Jacket Water, for Internal Combustion Engines, Arrangement for.

PUBLICATIONS

NAVSHIPS
0901-412-0002 - Diesel Engine Propulsion General Purpose.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity *or* as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 222 South Riverside Plaza, Room 1106, Chicago, Illinois 60606).

NATIONAL CLASSIFICATION BOARD
National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc. 1616 P. St. N.W., Washington, D.C. 20036).

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards *for* Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Diesel generator set.

3.1.1 Design. The requirements specified herein shall be considered as minimum standards and the design shall have as primary objectives the following:

(a) Maximum reliability.
(b) Maximum ease of operation.
(c) Maximum maintainability.

3.1.2 Reliability and maintainability. The contractor shall employ in the process of design and manufacture such methods as are necessary to assure achievement of levels of reliability and maintainability of the complete generator set not less than those specified in 3.1.2.2 and 3.1.2.5 herein.

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3.1.2.1 Reliability and maintainability program and plan. The contractor shall develop, submit for review/maintain and implement a reliability and maintainability program and plan using MIL-STD-785 (see 6.2) and MIL-STD-470 (see 6.2) as guides. Where a program plan and report for similar equipment was previously submitted by the contractor, the contractor need only propose any changes considered appropriate because of dissimilar items and current intended use.

3.1.2.1.1 The reliability and maintainability program and plan shall cover as a minimum control, monitoring, and conformance to 3.1.2.2 through 3.1.2.5.

3.1.2.2 The diesel generator set design shall be based on a life expectancy of 125,000 hours of operation with no limitations as to the number of starts and stops. This includes 100,000 hours at rated load, 20,000 hours between 50 percent and 100 percent rated load, and 5000 hours below 50 percent rated load. The time between overhauls shall be not less than 10,000 hours.

3.1.2.3 The design shall be such that parts which are not classified as onboard repair parts need not be repaired or replaced except during overhaul periods or where damage is incurred due to external causes unrelated to the design.

3.1.2.4 The design shall be such as to minimize the need for replacement of parts which are in the onboard repair parts category between overhaul periods. Such features shall be predicated on the following:

(a) Generator sets are installed and operated in accordance with the instructions contained in the technical manual.
(b) Planned maintenance actions are accomplished by ships force in accordance with the manufacturer's recommendations as developed from the maintenance engineering analysis (MEA) conducted in accordance with MIL-M-24365.
(c) Recommended inspections and repair actions during overhaul are accomplished in accordance with instructions and tasks developed from the MEA and when necessary, complementing information contained in the technical manuals.
(d) Generator sets are expected to operate for 10,000 hours between overhaul periods.
(e) The loads involved during operation between overhaul periods will be as specified in 3.1.2.2.

3.1.2.5 Reliability prediction. The reliability prediction of the generator set shall be based on the failure rate of parts, and the information specified in 3.1.2.1 through 3.1.2.4. The design and application of parts shall be such that the Mean-Time-Between Failure (MTBF) of the generator set is at least 1000 hours and the mean-time-to-repair (MTTR) is no more than 24 hours. The MTBF shall be predicted using the design prediction procedure of MIL-STD-756 and using the failure rate sources specified therein. For parts where no failure rate is available, a failure rate shall be estimated and the basis for the estimate stated in the calculations. The derating factors used for each part and the stress levels at which each part is operating shall be stated in reliability calculations. The stress levels shall be those as determined from test insofar as practicable. In those instances where test data is not available, design date shall be used.

3.1.2.6 A failure is defined as occurring when the collective or individual unit performance degrades to the point where the performance requirements of this specification cannot be met or wherein personnel safety is jeopardized.

3.1.2.7 The maintainability of the complete generator set shall be predicted in accordance with procedure IV of MIL-HDBK-472. The failure rates used shall be those which have been determined in the reliability predictions. The maintainability prediction shall take into consideration all design changes made during the engineering design phase.

3.1.2.8 Failure modes and effects analysis. The contractor shall conduct before the design tests a failure modes and effects analysis (FMEA). Prior to conducting the FMEA, the contractor shall prepare an FMEA selection report to describe the depth of the analysis. The content of the report shall be as specified in the contract or order and shall be submitted for contracting agency review. The analysis shall include as a minimum the critical subsystems, the accessories, controls, and parts delivered under the contract whose failure would cause the system to fail. The FMEA shall include the identification and tabulation of all ways (or modes) in which any part can fail and the effect of each failure mode on the next higher assembly and the next higher assembly and the system as a whole,

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The FMEA shall include, but not be limited to, the following:

(a) List of critical assemblies, subassemblies or parts which, by their failure rates, contribute the most to the unreliability of the generator set.
(b) Failure modes of the assemblies, subassemblies, or parts and corresponding probability of failure.
(c) Effect of each failure on the equipment and the interfacing systems, and the effect of interfaces on the equipment.
(d) Possible means of reducing probability of critical failure modes.
(e) Details to assist in preparation of the trouble-shooting section of the technical manual.

3.1.2.9 Maintenance engineering analysis. The contractor shall conduct before the design tests a maintenance engineering analysls (MEA). The MEA shall be conducted for each functional system, equipment, unit, assembly, or subassembly which requires maintenance, and the continued operation of which is essential to the successful completion of a mission itself. The MEA shall include as a minimum the MEA revision data (block 16) and the MEA summary (block 17) of worksheet 1, and fully completed worksheets IV and V of MIL-M-24365. It shall reflect the characteristics and design of the equipment delivered as production units.

3.1.2.10 Reliability and maintainability (R and M) data collection and reports. The contractor shall conduct an Rand M data collection , analysis, and corrective action program throughout all phases of testing. When a failure is experienced or discovered during- factory examination and tests a failure analysis as defined in MIL-STD-721 shall be provided. The program and related reports shall contain as a minimum the following elements:

(a) R and M prediction reports.
(b) FMEA report.
(c) MEA report.
(d) A documentation of all failures, corrective maintenance actions, and technical problems relating to reliability and maintainability.
(e) An analysis of each failure and technical problem.
(f) The determination, implementation, verification, of the results of corrective action.
(g) Maintainability demonstration test report.
(h) Reliability demonstration test report.

3.1.2.11 Design for maintenance.

3.1.2.11.1 The design shall provide accessibility to all parts which require inspection, preventative maintenance, or replacement in service with the minimum practicable need for disconnection or removal *of* another part or assembly. Access panels, covers, peep holes, etc. shall be employed to the fullest extent feasible and compatible with the intended inspection or maintenance action. The envelope of the minimum required space around the generator set which is needed for ease of maintenance shall be shown on the generator set installation drawing.

3.1.2.11.2 All parts, including repair parts furnished under the same contract or order or manufactured to the same drawings shall be interchangeable without the necessity for further machining, selective assembly or hand fitting of any kind to the fullest extent feasible.

3.1.3 System Safety program. The contractor shall develop and maintain an effective system safety program that is planned and integrated into all phases of design, production, and testing of the equipment. The system safety program shall provide a disciplined approach to identify hazards and prescribe corrective actions in a timely cost effective manner. The system safety program tasks shall be specified in a formal plan (System Safety Program Plan). The plan shall include requirements to be imposed on each subcontractor to assure compatibility with the system safety program for the equipment. MIL-STD-882 shall be used as guidance for preparing the System Safety Program Plan (SSPP).

3.1.3.1 Safety analysis. Safety analyses shall be performed to identify hazardous condition for the purpose of their elimination or control. Safety analyses shall examine the equipment, subsystem, components and their interrelationship including logistic support, training, maintenance and operational environments.

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3.1.4 The diesel generator set shall consist of the diesel engine, generator and such attached and unattached accessories as to make a complete generator set (when installed on shipbuilder-furnished foundations) capable of functioning in accordance with this specification without dependence on any other ship auxiliary when connected to the ship's fuel, exhaust, electric and water systems. Generator set hold down bolt holes provided for securing the equipment to the ship's foundation shall be designed for use with non-fitted bolts on one side and fitted bolts on the other in order to ensure there are no adverse effects on operation or alignment as a result of hull deflections.

3.1.5 Rating and characteristics. The rating and characteristics of the generator set shall be the follows:

Rated capacity	KW ratings for surface and snorkel shall be as specified (see 6.1)
Speed	As specified (see 6.1)
Voltage	450 volts a.c.
Frequency	60 Hertz
Phases	Three
Excitation	Static
Power factor (p.f.)	0.8 lagging
Ambient temperature	50°C.
Generator enclosure	Water-air-cooled
Bearing	Sleeve, forced lubricated from engine
Insulation	Class B
Starting	A i r
Voltage regulation	In accordance with MIL-R-2729 type III (no standby required)

3.1.5.1 The diesel generator set shall be capable of continuous operation at the rated snorkel and surface loads as specified (see 6.1) under conditions of inlet vacuum and exhaust back pressure as specified (see 6.1). The generator set shall also be capable of the following:

(a) Under snorkel and surface conditions with a base load on the set as specified (see 6.1), start a hydraulic pump motor of horsepower (H.P.) rating, inrush current and p.f. as specified (see 6.1), and maintain generator terminal voltage within plus or minus 18 percent of initial terminal voltage. The terminal voltage shall return to and remain within plus or minus 3 percent of the final sustained voltage within 2 seconds of the initiation of the start.

(b) Start a motor generator set of KW rating and inrush current as specified (see 6.1) with no other load on the set.

3.1.6 The diesel generator set shall be designed for automatic parallel operation and division of real and reaction power. The set, however, is not intended for automatic parallel operation but will be required to operate in parallel under manual or automatic control for short periods for the purpose of transferring real and reactive power from or to the ship's turbine generator sets, motor generator sets, or shore power connections. The necessary governor and voltage regulator adjustments shall be provided and adjusted so that operation during such transfer of power shall be stable and controllable under any required operating conditions.

3.1.7 The diesel generator set and accessories shall be as lightweight and compact as possible consistent with the design, performance, and maintainability requirements specified herein. If weight and dimensional constrictions are required, they shall be as specified (see 6.1).

3.1.8 List, pitch, roll and trim. The diesel generator set and associated auxiliaries shall be designed to operate in accordance with the performance requirements of this specification, and to avoid loss of oil under the most unfavorable combination of the following conditions of trim, pitch, list, and roll:

Trim	Plus or minus 7 degrees
Pitching	Plus or minus 10 degrees
Time of complete pitch cycle	5-10 seconds
List	Plus or minus 15 degrees
Rolling	30-60 degree
Time of complete roll cycle	8-15 seconds

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In addition, the set shall not leak oil under trim angles up to 30 degrees. This is not an operating condition. Degrees indicated are measured from the vertical to either side for list and roll, and from the horizontal fore and aft plane, up or down, for trim and pitch. The conditions of permanent list and roll or trim and pitch shall not be considered additive. The resultant angle of inclination from the following combinations shall be met.

(a) List and trim
(b) List and pitch
(c) Roll and trim
(d) Roll and pitch

3.1.9 The contractor shall provide the following items:

(a) A low engine speed shutdown control for snorkel operation which shall include a shutdown device, switch, and governor solenoid operating from a 115 volt, 60 Hertz, a.c. source. The device shall be adjustable between approximately 50 percent to 90 percent of rated speed, and shall actuate the switch when the speed falls to a minimum of 80 percent of rated speed. The switch shall reenergize the governor solenoid which in turn shall shut the engine down. The governor solenoid shall be provided with a manual over-ride for emergency stops and for set starting with the solenoid deenergized. The low speed switch shall be of the double pole, single throw type, 3 to 5 pounds per square inch (p.s.i.) range, set to open on decrease of lube oil pressure. An identical low speed switch shall also be installed to trip the generator circuit breaker when the speed falls to a minimum of 80 percent of rated speed. The settings of these switches shall be such that the generator breaker switch operates first.
(b) An overspeed trip actuated switch for tripping the generator circuit breaker.
(c) A 1/2 inch connection in the blower inlet and discharge for installation of a differential temperature thermometer.
(d) A 1/2 inch connection in the engine crankcase for manometer connection.
(e) A governor synchronizing motor, single phase, 115 volt a.c., readily reversible.

3.1.10 Shock. The diesel gnerator set and associated accessories shall meet the shock requirements of the appendix.

3.1.11 Speed characteristics. The speed governing system shall be of the speed sensing mechanical-hydraulic speed characteristics as defined by MIL-STD-178 as follows:

(a) Steady state speed regulation limits (percent) . . . . . 3.6 to 4.0
(b) Maximum steady-state governing speed band (percent plus or minus of rated speed) 0.33
(c) Minimum range of speed regulation changer (percent of rated speed) . . . . . 0.0 to 5.0
(d) Minimum range of speed changer (percent of rated speed) . . 90 to 105
(e) Maximum permissible momentary underspeed and overspeed (percent of rated speed) upon application or removal of 100 percent load 7.0
(f) Maximum permissible recovery time in seconds following underspeed and overspeed 5
(g) Prescribed speed band for determination of recovery time (percent plus and minus of rated speed) 0.5

3.1.12 Short circuit. The generator set shall be capable of withstanding without damage to any parts, the torques associated with a 3 phase fully asymmetrical short circuit at the switchboard assuming 25 feet of interconnecting cables from the generator terminals to the switchboard. (Also see 3.3.6.)

3.1.13 Vibration resistance. The diesel generator set shall meet the vibration requirements of types I and III in accordance with MIL-STD-167.

3.1.14 The generator set shall be provided with an air starting system as specified in 3.2.5.7. The generator set shall start, continue operating and be capable of providing rated voltage at the generator terminal within 10 seconds after the starting mechanism is set in operation without the use *of* cold starting aids. The set shall be capable of providing full output power immediately after completion of the start.

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3.2 ENGINE REQUIREMENTS

3.2.1 Engine rating. The engine h.p rating and speed shall be indicated on the engine identification plate (see 3.2.7.1). This rated h.p.output corresponds to full power rating of the generator set. This engine rated hp. shall be not greater than the hp. value chosen as the 100 percent "rated load" condition for the first unit endurance test specified in 4.5.2.

3.2.1.1 It is the intent of this specification to permit procurement of current production engine models with a demonstrated background of service experience. In addition to meeting all other requirements of this specification that are applied by the procurement document, a production engine at the time of the bid must have met at least one of the following requirements:

(a) The engine is essentially identical to an engine which has been used in Naval service at the same or a higher power rating and brake mean effective pressure (bmep).
(b) The engine shall be a type which the manufacturer will certify and has a background *of* 5000 hours operation on at least one unit. The unit or units considered as verifying this experience factor shall be installed for a rated output at least equal to the rating specified in the procurement involved. The engine supplier shall on specific request, provide detailed information as to where the units are located, who operates them, and insofar as practical, how they *are* used. This information shall be used when desired to verify the manufacturer's stated compliance with this experience requirement.

3.2.2 Materials Corrosion resistant materials shall be used in salt water system piping and components. Coated or painted cast iron or other ferrous metals are not acceptable in these areas.

3.2.2.1 Dissimilar metals. Contact between dissimilar metals shall be avoided in order to minimize galvanic corrsion. An interposing material which will minimize or eliminate this galvanic effect may be used.

3.2.2.2 Aluminum parts. Aluminum parts, other than pistons, blowers and fuel pumps shall be protected from corrosion by one of the following methods:

(a) Exterior parts to be painted shall be anodized or treated with corrosion resistant chemical film.
(b) Exterior parts to be left unpainted shall be anodized.
(c) Interior parts in contact with oil do not require treatment.
(d) Inferior parts not in contact with oil shall be anodized or treated with corrosion resistant chemical film.

3.2.2.3 Threaded holes in aluminum intended for bolts or screws for components or accessories normally removed or disassembled during routine maintenance or examination shall have steel inserts. Where insufficient basic metal exists, studs can be used without inserts. Dimension, including thread data and length, and complete instructions covering the removal and replacement of all inserts used shall be included in the engine instruction manual.

3.2.2.4 Magnesium alloys. Magnesium alloys shall not be used.

3.2.2.5 Coatings. No paint, plastic or zinc coatings shall be applied to the interior of gear covers, lubricating oil sumps or any other interior surface that will be in contact with lubricating, fuel, or hydraulic oils.

3.2.3 Design.

3.2.3.1 Standard operating condition. The following operating conditions shall be used for the de-d testing of the engine and accessories:

Ambient air temperatures	100°F.
Barometric air pressure (dry) inches mercury (Hg)	29.92
Fresh water from engine (minimum) to 185°F. (maximum)	160°F.
Exhaust back pressure	3 p.s.i.
Sea water inlet temperature	90°F.
Sea water outlet temperature (maximum)	130°F.
Lubricating oil temperature from engine	160°F. (minimum) to 190°F. (maximum)
Intake air pressure	1 inch Hg. (vacuum)

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3.2.3.2 Inclination The engine, accessory components and piping systems shall be capable of operating in accordance with all the requirements of this specification and without loss of fluids, loss of lubricating oil pump suction, and without the connecting rod and gears dipping into the oil under the conditions specified in 3.1.8.

3.2.3.3 Thread fastening. All threaded hardware parts shall conform to Handbook H28 Nuts and other threaded fastenings on the engine and accessories shall be secured by wire, lockwashers, cotter pins, clips or other means, in such manner as to positively prevent the part working loose under operating conditions. All drain plugs shall be readily removable with general purpose wrenches.

3.2.3.4 Accessibility. Engine attached accessories such as pumps, blowers, coolers, filters, strainers and silencers, shall be designed and secured in such a manner as to permit maximum accessibility to the accessories and the engine. Engines shall be provided with adequate openings with removal covers in the frame and crankcase above the mounting flange for servicing and examining the engine.

3.2.3.5 Timing marks. Means shall be provided to permit checking of valve and injection pump timing. Means for timing and barring the engine over by hand shall be incorporated in the engine design.

3.2.3.6 Crackshaft. Nuts on coupling flange bolts shall be accessible and locked or keyed in place. Provisions shall be made on the free end of the crankshaft for attaching a torsiograph drive.

3.2.3.7 Shock resistance. The engine shall be designed to meet the requirements of the Appendix.

3.2.4 Construction.

3.2.4.1 Accessory equipment. The following accessories shall be furnished with each engine:

(a) Sea water pump
(b) Fresh water pump
(c) Lubricating oil pressure pump
(d) Lubricating oil priming pump
(e) Fuel supply pump
(f) Fuel priming pump
(g) Lubricating oil cooler
(h) Fresh water/sea water heat exchanger
(i) Lubricating oil filter(s)
(j) Lubricating oil strainer
(k) Intake air filter
(1) Fuel oil filter
(m) Fuel oil strainer
(n) Temperature regulator or thermostat
(o) Governor
(p) Overspeed trip device
(q) Starting system
(r) Blower
(s) Engine mounted controls
(t) Lubricating oil low-pressure alarm switch
(u) Fresh water high-temperature alarm switch
(v) Cylinder test and indicator cocks
(w) Engine turning or jacking device
(x) Remote shutdown system
(y) Instruments and panel
(z) Emergency shutdown device
(aa) Pyrometer and thermocouples

Other accessories or engine attached piping necessary for engine operation but which are not specified herein or in the contract or order shall be furnished by the engine manufacturer. Unless otherwise specified, piping, including fitting and flanges, alarm systems, wiring, or mechanical linkage required to connect detached accessories, instruments, and controls to the engine, or to cooling water, fuel starting air, electrical, exhaust, or lubrication systems, will be furnished by the installing activity.

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3.2.4.2 Gaurds and shields. Guards and shields shall be furnished as specified in 3.2.4.2.1 and 3.4.2.2.

3.2.4.1 Gaurds. Guards for protection of personnel shall be provided for flywheels and other exposed moving parts. Where there are no feasible means for attaching the guard to the equipment, the guards will be furnished by the engine builder and the installing activity will accomplish the installation.

3.2.4.2. Shields. All hot machinery surfaces shall be shielded or insulated so that no external surface the machinery shall exceed a temperature of 400°F. All fuel and lubricating oil lines or fittings, including gage lines, shall be located or shielded so that any leaks cannot drip or spray on any component with a surface temperature of 400°F. or higher.

3.2.4.3 Crankcase fumes. The engine builder shall provide a system to dissipate through the engine intake or exhaust system all oil fumes generated by the engine in the crankcase and other areas.

3.2.4.4 Engine crankcase and air box explosion protective devices or design. All diesel engines with cyclinder bore diameter of inches or greater shall protected from the hazard of crankcase or air receiver explosions by one of the procedures specified in 3.2.4.4.1 through 3.2.4.4.5.

3.2.4.4.1 If the crankcase can be demonstrated by test to withstand an internal gas pressure of 100 pounds per square inch gage (psig) without rupture, it is considered to be acceptably safe to contain a crankcase explosion and no other protection devices is required. For the purpose of this demonstration, the term crankcase shall be interpreted to include the entire engine structure that would be exposed to internal gas pressure in the event of a crankcase explosion.

3.2.4.4.2 If the integrity of the engine crankcase and related volume cannot be demonstrated as specified in 3.2,4.4,1, the engine shall be provided with spring loaded explosion relief valves in each cylinder crankpit, to provide a total relief area of at least 1-1/2 square inches, for each cubic foot of total crankcase volume. These relief valves shall be designed to be fully open at a crankcase pressure of 20 psig, and to close and seal the crankcase from air when the pressure has been relieved. The entire crankcase system in the terms of 3.2.4.4.1, including any access covers, shall be designed to withstand an internal pressure of at least 25 psig without failure when explosion protection devices are used.

3.2.4.4.3 The air receivers of two cycle engines (only) of 6 inch diameter or larger cylinder sizes, shall also be protected from air receiver explosions. If the air receivers, including all associated structure that would be subject to gas pressure in event of air explosion, can be demonstrated to withstand 125 psig internal gas pressure, no other air receiver explosion protection device is required.

3.2.4.4.4 If the pressure integrity requirements of 3.2.4.4 cannot be met, the air receiver shall be provided with an explosion relief valve or valves, having at least 1-1/2 square inches of relief area per cubic foot of total air receiver system volume. These relief valves shall be designed to be tight at maximum air receiver operating pressure and to be fully open at 20 psig pressure above that operating pressure level. The valves shall be designed to relieve the explosion and reset tightly. When these explosion relief valves are used, the entire receiver structure, covers, etc., which would be subject *to* the pressure of an air receiver explosion shall be designed to withstand at least 5 psig above the pressure at which the relief valve is fully opened.

3.2.4.4.5 Explosion relief valve design requirement. All crackcase or air box explosion relief valve, if used, shall be provided with blast deflectors which will direct explosion forces away from personnel who might be alongside the machinery.

3.2.4.5 Welding. Welding and allied processes shall be in accordance with MIL-STD-278.

3.2.5 Fluid piping systems.

3.2.5.1 Piping. Engine attached piping shall be strapped and supported by the engine structure to prevent vibration and resist shock. Takedown joints in fuel systems shall be kept *to* a minimum to reduce leaks. All connections from the engine to the ship's systems (i.e. exhaust, lubricating and fuel oil inlet and return, and the raw water supply and discharge) shall be in accordance with MIL-STD-438. Flange connections shall be Navy type. Navy flanges for steel or nonferrous piping and fittings shall conform to MIL-F-20670 or MIL-F-20042, as applicable. Except as exempted in 3.2.5.1.1, taper pipe threads are not permitted in engine piping systems, accessories or driven equipment. Thread connections, where used, shall employ straight threads in accordance with MS16142.

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3.2.5.1.1 Taper pipe threads are permitted in the following areas:

(a) Pipe plugs in sizes of 3/4 inch and below used for applications where design pressures do not exceed 50 p.s.i. Plugs shall be seal welded or brazed, if possible.
(b) Instrumentation, controls, vent, filling and drain connections where design pressures do not exceed 50 p.s.i. and where fluids handled; (1) are not toxic, (2) are not dangerous, (3) would not cause atmospheric contamination, (4) would not create a fire hazard, and which would not cause, in the event of failure, a major breakdown of equipment nor create a hazard to the surrounding area nor affect the operation of other vital equipment.
(c) Pipe plugs internal to the engine.

3.2.5.2 Cooling water system. Engines shall be designed to operate with a closed fresh water cooling system in accordance with Drawing 5000-S4501-64492. Sea water piping shall be of 70-30 copper nickel alloy. Fittings and valves shall be bronze. All parts of the sea water system directly connected with the sea which could be subjected to submarine submergence pressure shall be designed for a test pressure that will be furnished by the contracting officer upon request. Such parts shall be given radiographic inspection as specified by the contracting officer.

3.2.5.2.1 Cooling water pump. Fresh and sea water pumps shall be of the centrifugal type driven by the engine with the drive arrangement designed to prevent water leakage into the engine lubricating oil system. Where practical, the fresh and sea water pumps shall be interchangeable. The pumps shall be so mounted or provided with a continuous vent to prevent the entrapment of air. A self-priming seawater pump shall be provided where pump lift requirements demand such a pump. All pumps which are not self draining shall be provided with petcocks or other means of draining completely without the use of special tools. Special tools are defined as those tools not listed in the Federal Supply catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)) . Engine manuals shall contain complete instructions covering draining of pumps. Seawater pump volute and suction head shall be made of one of the following materials: composition M in accordance with MIL-B-16541; alloy D5 in accordance with QQ-C-390; copper-nickel alloy (70-30) in accordance with MIL-C-20159; or nickel aluminum bronze in accordance with MIL-B-23921. Shaft seals for sea water pumps shall prevent any air irom leaking into the pumps under a suction head of 15 inches of Hg. Sea water pumps shall deliver rated capacity when operating under a discharge head of at least 35 p.s.i.

3.2.5.2.2 Heat exchangers. Fresh water/sea water heat exchangers shall conform to type A, class 1 of MIL-C-15730.

3.2.5.2.3 Water temperature control. Temperature regulator valves and thermostatic valve shall be either the fail safe type which will pass all the cooling water through the cooler in the event of failure of the thermostatic element or be provided with a manual "over-ride" for the thermostatic element.

3.2.5.2.4 Inhibitors. Engines shall employ coolant inhibitors conforming to O-S-588 (chromate type).

3.2.5.2.5 Gaskets and packing. Gaskets and packing used in the cooling system shall be of materials resistant to deterioration by sea water or when any of the following are added to the fresh water system:

(a) Antifreeze compounds conforming to O-A-548.
(b) Corrosion inhibitor compounds conforming to O-S-588, MIL-I-24453 or 0-1-490.
(c) Preservative compounds conforming to MIL-C-16173 and MIL-L-21260.

3.2.5.3 Fuel system. Engines shall be capable of operating on fuel corresponding to both MIL-F-1684 or JP-5 of MIL-T-5624. The fuel system shall include the following:

(a) Positive displacement type fuel pump.
(b) Relief valve connected to discharge into pump suction or supply tank.
(c) Priming pump.
(d) Filters.
(e) Necessary piping, valves and fittings.

3.2.5.3.1 If leak-off from the injectors and fuel pump exceeds 2 percent of the supply pump capacity, necessary equipment shall be provided to automatically return leak-off fuel to the service tank.

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3.2.5.3.2 Fuel pump. The engine driven positive displacement fuel pump shall be capable of lifting fuel up to 48 inches.

3.2.5.3.3 Filtration. The fuel system shall be protected by a primary filter (strainer) ahead of the engine mounted fuel pump and a duplex secondary filter ahead of the fuel injection pump or unit injectors. If the fuel system is of such a design that the filter must be positioned on the suction side of the fuel pump, only the secondary filter is required. Filters shall be constructed of brass, bronze, or other metals which are not readily corroded by sea water contaminated fuels. Coated steel or untreated aluminum are not acceptable. Filters shall include vents to permit the expulsion of air. Manually operated drain valves shall be provided to completely drain the filter bodies without the use of tools.

3.2.5.3.3.1 Primary filter Primary filters (strainers) shall be the simplex metal edge type with 0.003 to 0.0035inch spacing, enclosed manually operated cleaning device and sediment drain. The term “metal edge” is defined as stacked metal discs or helically wound flat metal wire or metal ribbon media. Wire cloth, screen, sintered metal or perforated plate media are not acceptable for simplex strainers.

3.2.5.3.2 Duplex secondary filters. Duplex filters shall consist of two filter units connected to a common head equipped a three way plug valve and disposable filter elements. The level operated valve shall direct the fuel flow through one side to permit element change and cleaning of the other side without interruption of flow. Each side shall be sized to accommodate the full flow of the engine fuel pump and employ drain valves to permit complete draining. Filter elements and materials used for metal parts and gaskets shall be in accordance with MIL-F-20627. Filter elements shall be rated at 150 gph in lieu of the 50 gph rating stated in MIL-F-20627.

3.2.5.3.4 Fuel oil priming pump. A fuel oil pump to prime the engine fuel oil system within 2 minutes at an ambient temperature of 70°F. shall be furnished.

3.2.5.3.5 Fuel oil tank. The preferred Navy installation can be expected to provide, by the use of a day or other arrangement, a positive head of fuel pressure on the engine primary pump suction connection. The fuel systems of some engines may be so designed that combustion space flooding would result when this positive head is applied during engine shutdown. In such cases, the engine supplier shall furnish all necessary devices, such as float tanks, valves, etc., which he recommends to protect the engine and insure proper operation.

3.2.5.4 Lubricating system. Lubricating oil shall conform to symbol 9250 *of* MIL-L-9000 for ambient condition above 20°F. The same grade or symbol number of lube oil shall be used in the engine and attached accessories. Engines shall be provided with an oil level indicator, filling opening and accessible sump drain connection or a sump pump to drain or clean the engine oil pan. Bayonet type oil level indicators shall be shielded to permit accurate readings while the engine is running. The lubricating oil system shall be arranged as shown on figure 1. Engines shall be provided with suction and pressure connections for priming and flushing the engine lubrication system.

3.2.5.4.1 Lubricating oil pumps.Pressure pumps, scavenging pumps and piston cooling oil pumps shall be of positive displacement type, driven by the engine. The scavenging pump capacity shall be approximately 1-1/4 times the capacity of the pressure pump and shall scavenge the system from idle to rated speed. A relief valve shall be provided for the outlet *of* each pump. If the pump suction incorporates a check valve, means shall be provided to prevent excessive pressure build-up in the suction line in case the engine and pump are rotated backwards.

3.2.5.4.2 Coolers. Lubricating oil coolers shall conform to MIL-C-15730.

3.2.5.4.3 Lubricating oil filters. Engines shall be designed to use full flow filter elements conforming to type IV OF F-F-351 and MS35802. Full flow filter relief valves shall be external of the filter case. The valves shall be designed to bypass oil when the pressure drop is between 20 and 25 p.s.i. The valve closing pressure shall be not less than 20 p.s.i. The filter inlet shall be internally baffled or so designed to prevent the oil flow from impinging directly on the filter elements. The filter shall include a hand operated drain valve which shall completely drain the filter for servicing.

3.2.5.4.4 Lubricating oil stainers. Strainers shall be the simplex metal edge type With 0.005 inch spacing and enclosed manually operated cleaning device. The term “metal edge” is defined as stacked metal discs or helically wound flat metal wire or metal ribbon media. Wire cloth, screen, sintered metal or perforated plate media are not acceptable under this specification. The strainer shall be sized to carry the full flow *of* 150 Saybolt Seconds Universal (SSU) viscosity oil with a pressure drop of not more than 5 p.s.i. with

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clean elements. The strainer bypass valve shall be set to open within the range of 20 to 23 p.s.i. differential pressure across the element, and shall be capable of passing the full oil flow with a total pressure drop through the bypass of less than 26 p.s.i. Strainers shall include hand operated drain valves which shall permit complete drainage of the body(s) for servicing.

3.2.5.4.4.1 Lubricating oil priming pump. A lubricating oil pump to prime the engine lubricating oil system Within minute at an ambient temperature of 70°F. shall be furnished.

3.2.5 Air intake system. The engine will take combustion air from shipbuilder furnished ducting directly to the engine intake. An air filter shall be installed in the ducting as close to the engine as possible.

3.2.5.5.1 Intake air filter. An air filter using filter media in accordance with MIL-F-7194 shall be furnished with each engine. The filter elements shall be designed to withstand a uniformly distributed pressure of 5 p.s.i. over the entire surface of the element. The area of the filter element shall be such that the air velocity at the face of the filter does not exceed 2000 feet per minute, or the pressure drop across the clean filter be more than 4 inches of water at rated output. A manometer or signal device shall be connected across the element to indicate when cleaning is required. The filter housing shall permit removal of the element for cleaning. The filter element shall be installed at an angle 90 degrees or less to the direction of air flow in order that impinging dirt will drop to the bottom of the duct on the upstream side of filter element.

3.2.5.5.2 Blowers and air recievers. Means shall be provided for draining pockets in the blower housing and air receiver where oil or water may accumulate.

3.2.5.5.6 Exhaust system. Exhaust headers shall be fresh water cooled, insulated or shielded. Provisions shall be made for draining the gas spaces of the header from either end.

3.2.5.6.1 Exhaust emission. The engine shall be designed to keep exhaust smoke emission to a minimum. The Ringlemann smoke chart shall be the standard of comparison. During normal, constant operation at rated speed the smoke opacity shall not exceed Ringlemann number 1. During a speed change, smoke opacity may exceed Ringlemann number 2 for 5 seconds, then gradually return to less than Ringlemann number 1 in less than 30 seconds.

3.2.5.7 Air starting. The air starting system shall consist of the starting equipment including motor or distributor and applicable valves, strainer, lubricator, lines, connections and fittings. The ships air supply will be at the pressure specified (see 6.1). Any necessary air reducing valves shall be furnished on the basis of one valve for each engine room. The system shall include a device to automatically shut off the starting air when the engine starts.

3.2.5.8 Speed governing system. Engines shall be equipped with a speed governing system and an independently driven over-speed device. Speed regulating governors shall be as specified in 3.1.11.

3.2.5.8.1 Overspeed protection. As protection against dangerous engine overspeed due to failure of the regulatlng governor, its linkage or other engine malfunction, each engine shall be equipped with a separate and ‘independent speed limiting device that will prevent the engine from exceeding rated speed by more than 15 percent by cutting off completely the fuel to the engine. This device shall require manual resetting.

3.2.5.9 Instrumentation. Instruments shall be furnished with each engine for the following measurements:

(a) Pressure indicators:
(1) Fresh water discharge.
(2) Differential type gage indicating the sea water pressure to and from the pump suitable for the required submergence pressure (see 6.1) .
(3) Duplex gage to indicate the lubricating oil pressure at the pump discharge and at the full flow filter outlet.
(4) Piston cooling oil (if separate pumps are used).
(5) Duplex gage indicating the lubricating oil pressure to and from the strainer.
(6) Duplex gage indicating fuel oil pressure to and from the filter.
(7) Starting air at the engine.
(8) Scavenging air pressure.
(9) Lubricating oil pressure at the turbo charger or blower.

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(b) Distant reading temperature indicators:
(1) Lubricating oil from engine.
(2) Fresh water from engine.
(c) Temperature indicators shall be furnished by
lation by the installing activity for the fo
(1) Sea water to and from heat exchanger.
(2) Fresh water to lubricating oil cooler.
(3) Fresh water to the engine.
(4) Lubricating oil to engine.
(d) Alarms :
(1) Low lubricating oil pressure.
(2) High temperature cooling water.
(e) Electrical temperature pyrometer and selector switch:
(1) Each cylinder exhaust outlet.
(f) Tachometer:
(1) Engine speed.
(g) Time indicator or hour meter:
(1) Engine operating time.

3.2.5.9.1 Panal. An instruction panal contained all instructments listed in 3.2.5.9(a), (b), (c), (f), (g) and 3.3.9 shall be furnished with each engine designed for mounting on the engine, off of but adjacent to the engine or located at the engine control stand. Panal mounted instruments shall be flush mounted and 3-1/2 or 4-1/2 dial size and shall be shock resistant design. All duplex gages shall be 4-1/2 inch size. Instruments for reading from operating station and for maintenance and calibration checks. Instruments shall be accordance with and installed inaccordance with table I as applicable:

Table I - Instruments.

Instrument	Specification	Remarks
Pressure instruments: Indicators Transducer Differential pressure transducer	MIL-I-18997 MIL-P-24212 MIL-D-24304	Installed in accordance with Drawing 810-1385850.
Alarm	MIL-S-16032	Alarm sensor shall be installed at the remote end of the lubricating oil system.
Temperature instruments: Direct reading (bimetallic) Remote reading (filled system) Electrical sensors Signal conditioner and power supply	MIL-I-17244 MIL-T-19646 MIL-T-24388 MIL-T-24387	----- ----- ----- -----
Thermowells	MIL-W-24270	Installation of thermowells shall be in accordance with Drawing 810-1385917.
Pyrometer (thermocouple and selector switch)	Type PM, MIL-P-16304	Cabling to be furnished by installing activity.
Alarm	MIL-S-16032	Sensor located in cooling water outlet.
Cable	MIL-C-915	-----
Rpm indicators 1/ Remote station indicator, when required	MIL-T-16049 Type IC/EFB, MIL-T-16049	Range shall be not less than 25 percent above rated engine speed.
Timer 1/	Totalizing 9999.9 hours electric, operates when engine is running.	Totalizes number of hours of engine operating time.

1/ Timer and tachometer may be combined.

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3.2.5.10 Control system. The engine shall be provided with attached control levers for starting stopping and speed regulation. Levers necessary for operation of the engine shall be conveniently grouped at one e.d of the engine. The governor fuel actuating shaft, or fuel injection pump control linkage shall be provided with a pointer and a graduated scale to indicate pump rack position.

3.2.5.10.1 Auxiliary hand operation. When remote operation is required, for controlling the engine speed auxiliary hand operation shall be provided. The design shall include means to prevent simultaneous power and hand operation. Hand or power operated control mechanism shall permit convenient operation by one man.

3.2.5.10.2 Emergency shutdown device. Engines shall be provided with an emergency shutdown device, operable by a pull cable, which shall trip the fuel racks. The device shall be of a type which-requires manual resetting before the engine can be restarted. This device may be combined with the speed limiting device (see 3.2.5.8.1). The pull cable, sheaths, guides, pulleys, handles and hardware shall not be constructed of any materials that would burn, melt or render the control inoperable in case of an engine room fire.

3.2.6 Painting. Engines shall be painted in the same manner as for commercial delivery. Internal oil-wetted surfaces shall not be painted.

3.2.7 Designating and marking. Engines, components and parts shall be marked for identification in accordance with MIL-STD-130

3.2.7.1 Engines shall be provided with an identification plate secured to the engine in a visible and convenient location. Identification plates shall be of metal and shall show the following:

(a) Engine type.
(b) Serial number.
(c) Federal stock number.
(d) Contract number.
(e) Model number.
(f) R.p.m.
(g) Bhp
(h) Rotation.
(i) Bore.
(j) Stroke.
(k) Dry weight.

3.2.7.2 Complete accessory components such as vendor furnished pumps, coolers, filters, governors, electrical and starting equipments shall be provided with identification plates showing manufacturer's name, model or part number, and capacities or ratings where applicable.

3.2.7.3 A plate with an arrow showing direction of rotation shall be secured to the engine at the power take-off end of the engine.

3.2.7.4 Repair parts shall be marked with the manufacturer's part number by casting, stamping, or etching. Where it is not practical to mark parts by any of these methods, the parts shall be identified by tags or printed description on the package.

3.3 A.c generator. The generator shall conform to MIL-G-3124, except as otherwise specified hereinafter, and shall have the following classification:

Type required	Type II
Insulation	Class B
Rating	
Voltage	450 volts a.c.
Phases.	Three
R.p.m.	As specified
Kilowatts (KW)	As specified (see 6.1)
Frequency	60 Hertz
Power factor (p.f.)	0.8 lagging
Ambient temperature	50°C.
Prime mover	Diesel engine
Enclosure.	Water-air-cooled
Bearing.	Sleeve, forced lubricated from prime mover
Excitation system	Static, in accordance with type III of MIL-R-2729

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3.3.1 Reduced weight and size shall be considered as major factors in the generator design. The generator shall be of single, sleeve type bearing and shall have the engine end of the shaft piloted in the engine crankshaft. The generator half of the coupling flanges shall be an integral part of the generator shaft forging. A sight flow fitting shall be provided at the bearing oil discharge.

3.3.2 The lower half (up to the bottom of the shaft) of the generator enclosure shall be watertight, and the upper half of the enclosure shall be spraytight.

3.3.3 The generator air cooler shall be in accordance with class 1 of MIL-C-19836 and shall be designed for the "submergence design pressure" and "submergence test pressure" specified (see 6.1). Cooling water shall be furnished from the engine seawater system.

3.3.4 The generator efficiency shall be not less than 88 percent at 50 percent load and 91.5 percent at 100 percent load.

3.3.5 Heaters shall be for 440 volt, three phase 60 Hertz operation, and shall be of the rating required.

3.3.6 The generator (and associated excitation and voltage regulation system) shall be capable of withstanding without injury to any part, a three phase or single phase short circuit at the terminals for a period of 30 seconds with the generator under automatic control of the voltage regulator. The sustained value of current during a three-phase short circuit shall be at least 3.2 times rated current. Total temperatures during these periods shall not exceed *safe* valves.

3.3.7 At rated frequency and at a p.f. of 0.5 or less lagging, the generator, when under control of its voltage regulator, shall be capable of delivering 150 percent rated current for 1 minute and maintaining the terminal voltage within 5 percent *of* rated voltage.

3.3.8 Where tapered keys or wedges are used in conjunction with dovetail laminations to secure the complete pole assembly to the rotor spider, means shall be provided to prevent their loosening due to shock or excessive vibration.

3.3.9 Requirements covering bearing thermostatic switch and temperature indicating equipment of MIL-G-3124 and air temperature alarm of MIL-C-19836 do not apply. In lieu thereof, continuous reading resistance type thermometers shall be provided for measuring the temperature of the air in and out of the cooler and the bearing, and shall be mounted on the engine gage board.

3.3.10 The location and arrangement of the generator main terminals, cooler connections, field rectifier terminals (If rectifiers are mounted in air stream and terminals are located outside generator enclosure) , and heater connections shall be as required by the installation. A mounting pad or flange shall be provided for the attachment of a shipbuilder furnished terminal box for the main terminals and rectifier terminals if required.

3.3.11 Damper windings shall be provided and shall be connected between poles.

3.3.12 Wave shape.

3.3.12.1 Deviation factor. The deviation factor of the open circuit terminal voltage shall not exceed 6 percent on any or all phases.

3.3.12.2 Harmonic content. From no load to full load, the r.m.s. value of the sum of all harmonic components in the output voltage wave shall not exceed 5 percent of the r.m.s. value of the rated output voltage. No single harmonic shall exceed 3 percent. Harmonics of 0.2 percent *or* less may be disregarded in the determination of the total harmonic content.

3.4 Excitation and voltage regulation system. The excitation and voltage regulation system shall be of the completely static type in accordance with type III of MIL-R-2729 except as otherwise specified herein.

3.4.1 Voltage regulator. The voltage regulator shall have the following classification:

(a) Type of control . . . . . Static
(b) Enclosure . . . . . Protected for switchboard mounting
(c) Regulators per switchboard . . One normal
(d) Operation . . . . . Parallel (see 3.1.6)

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3.4.1.1 All voltage regulator components and parts, except the manual control unit, the voltage adjusting unit, the control switch, field flashing switch, and droop compensation transformers shall be mounted in one box. The control units and switches shall be suitable for back of board mounting, front of board operation.

3.4.2 Static exciter components shall be suitable for mounting within the switchboard enclosure unless otherwise specified (see 6.1). Power rectifiers, if desired may be mounted in the generator cooling air stream provided they are readily accessible *for* examination and replacement.

3.4.3 Separate current, potential, and combination transformers shall be used with each regulator-exciter system and no other burden shall be placed on the transformers. Transformers shall be ungrounded.

3.4.4 Semi-conductor devices.

3.4.4.1 All semiconductor devices shall be selected from MIL-STD-701 and shall conform to the requirements of MIL-S-19500. Nonlisted devices shall require .specific acceptance by the command or agency concerned as required by MIL-E-917. All rectifiers shall be of the silicon type. All devices shall be individually replaceable.

3.4.5 Capacitors. Capacitors shall be selected and applied in accordance with MIL-E-917.

3.4.6 Resistors and rheostats. Resistors and rheostats shall be selected and applied in accordance with MIL-E917.

3.4.7 An external source of d.c. power will be available and may be used for field flashing of the a.c. generator. If field flashing is provided from an a.c. source through rectifiers, the flashing switch shall open both the a.c. and d.c. sides of the circuit in the “off” position.

3.4.8 A “De-energize” position shall be provided on the regulator control switch in addition to the “Auto” and “Manual” positions. This “De-energize” position shall operate in the control circuitry such as to result in no generator output voltage except from residual magnetism.

3.4.9 Duplicate but electrically isolated control power transformers (where used) shall be provided for the automatic and manual control circuits and shall be mounted adjacent to each other to permit back-up of one by the other by simple wiring changes.

3.4.10 Short circuit current. The excitation and voltage regulation system shall provide excitation to the generator under both single phase and three phase short circuit conditions. The excitation shall be such that, for a three phase short circuit, the sustained line current shall be at least 3.2 per unit. In addition, the system shall be capable of maintaining the short circuit current for a period of 30 seconds, and shall restore normal voltage upon removal of the short.

3.4.11 Generator overload. The excitation and voltage regulation system shall provide necessary excitation and control to maintain generator terminal voltage within 5 percent of rated voltage with the generator operating at 150 percent rated current at 0.5 p.f. lagging and rated frequency for 1 minute.

3.4.12 The excitation and voltage regulation system shall not contribute more than 0.5 percent to the generator deviation factor, nor more than 0.5 percent to any individual voltage harmonic.

3.4.13 Generator loads will normally be of lagging p.f. between 0.5 and 0.8. It is possible, however, that generators may occasionally operate at leading p.f. or greater lagging p.f. At rated KW load, at any p.f. from 0.8 lagging to 0.8 leading, the system must be capable of maintaining generator terminal voltage within one percent of the required normal 0.5 to 0.8 lagging p.f. steady state voltage regulation band.

3.4.14 The excitation and voltage regulation system shall meet the requirements for type I vibration of MIL-STD-167.

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3.4.15 An overvoltage limiting circuit shall be incorporated in the excitation and voltage regulation systen. This circuit shall automatically limit generator overvoltage to a value not exceeding the set value between 105 and 130 percent of rated voltage. The voltage limiting level shall be adjustable between 105 and 130 percent of rated voltage. The normal setting for overvoltage limiting shall be 110 percent of rated voltage.

3.5 Drawings.

3.5.1 Electrical. Drawings for the generator and excitation and voltage regulation system shall be furnished in accordance with MIL-D-1900/2, types II and III. Drawing use categories are A, B, D, G, H and I of MIL-D-1000. Contents of drawings shall be as specified in MIL-E-917, MIL-G-3124, and MIL-R-2729 as applicable.

3.5.2 Engine Drawings of production or existing parts, components, or equipment applicable to a contract or order shall consist of the manufacturer's commercial shop drawings. New or revised drawings shall be in accordance with MIL-D-1000/2, type II. Drawing use categories are A, B, D, G, H and I of MIL-D-1000. Drawings shall include all parts *of* the engine and accessories not exempted herein.

3.5.2.1 Accessory drawings. The following accessory drawings shall be furnished:

(a) A cross section drawing with bill of material for the following:
(1) Regulating and overspeed governors
(2) Cold start device
(3) Lube oil filter
(4) Thermostats or temperature regulators
(5) Starting system components
(6) Air intake filter/silencer
(7) Pneumatic control devices
(8) Fuel oil filter and strainer
(9) Mufflers
(10) Coolers
(11) Water filter

(b) An outline drawing for the following accessories:
(1) Tachometer
(2) Tachometer generators
(3) Pressure gages
(4) Thermometers
(5) Thermocouples
(6) Flexible exhaust hose
(7) Expansion tanks
(8) Valves, (relief, safety, throttling, regulating, reducing, and similar types) .

(c) Drawings of items on qualified products lists and standard hardware that are fully described in bills of material and drawings of pieces used in the fabrication of welded parts shall not be included.

3.5.3 Generator set installation drawing or drawing. Installation drawings for the combined diesel generator set shall be furnished in accordance with MIL-D-1000/2, type II and shall contain the following information. Drawing use categories are A, B, and G of MIL-D-1000:

(a) Dimensional outline drawings (four view drawings) of the generator set and each detached accessory. These drawings shall identify and show the location of shipbuilder connections, mechanical and electrical.
(b) Foundation bolting details.
(c) Piping system diagrams, showing pipe sizes, type of connections, relative location of coolers, valves, strainers, filters, thermometers, gages, alarm devices, and quantity, pressures, temperatures of oil, air and water.
(d) Data table as necessary to clearly describe all equipment, including capacities of pumps and blowers at rated speed, rated BHP, BMEP, RPM, type numbers, ratings (volts, amperes, rpm, KW, torque and so forth), name of manufacturer(s).
(e) Table of weights showing wet and dry weights of all major units weighing 100 pounds or more and of the complete generator set package.
(f) Centers of gravity of engine and generator and of the combined generator set in at least 2 planes.
(g) A sketch of the engine gage board showing mounting features and the location and identification of all gages.
(h) Clearance required to remove generator rotor, generator air cooler, and engine parts (see 3.1.2.11.1).

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(i) All openings for examination or clearance measurements.
(j) Maximum amount of heat in BTU per hour dissipated by the generator set into the space where installed.
(k) Location and size of lifting means.
(l) Any additional information required for the proper installation and operation of the equipment.

3.5.4 Preliminary drawing for review. The contractor shall submit to the ordering activity the required number of blueprint (or equivalent) copies of preliminary drawings, as specified (see 6.1) , for review. Drawings shall consist of all installation drawings and any other assembly or detail drawings specified (see 6.1).

3.5.5 Final drawings. Final drawings shall consist of microfilm of all installation, assembly and detail drawings required by this specification. Microfilm shall be mounted in aperture cards conforming to MIL-M-38761/2 and shall be of the type and quantities specified (see 6.1).

3.5.5.1 Tabulating index cards conforming to MIL-M-38761/2 shall be furnished as specified (see 6.1).

3.6 Manuals. Manuals shall conform to type I of MIL-M-15071. The following equipment shall be covered in the manual:

(a) Diesel engine(s) and associated accessories.
(b) Generator and air cooler.
(c) Excitation and voltage regulation system.
(d) Starting equipment.
(e) Temperature indicating and other alarm equipment.
(f) Parts lists.

3.6.1 In addition to the requirements of MIL-M-15071, the following information shall be included:

(a) Reduced size prints of the combined unit installation drawings and all final drawings for electrical equipment.
(b) Copies of the first unit test results for the generator, excitation and regulation system, and the combined diesel generator set.
(c) Complete and comprehensive alignment instructions for the entire plant. The instructions shall cover original alignment procedure, with maximum limits for radial and axial deviation, and information for alignment checks after installation. A machinery layout sketch showing locations of measurements, and space for recording measurements shall be provided.

3.6.2 Preliminary manuals. The preliminary drafts of the technical manual shall contain all the information which the generator set manufacturer proposes for the final manual except for photographs and results of tests. Preliminary manuals shall be submitted to NAVSHIPS for approval.

3.6.2.1 Preliminary drafts of the manual shall be submitted for review in the quantity on the date and to destination as specified in the contract.

3.6.3 Final manuals. Responsibility for preparing final manuals, quantity of manuals required, date manuals are required and distribution of final technical manuals shall be as specified in the contract or order.

3.7 Repair parts and special tools. Repair parts and special tools shall be furnished in accordance with the requirements of method A or B in MIL-P-15137 as specified (see 6.1). When method A is specified, onboard repair parts and tools shall be based on the number of identical generator sets installed per ship, as specified (see 6.1) and the following:

(a) A set of special tools which are required for disassembly, repair adjustment, and reassembly, shall be furnished on a per ship basis. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).
(b) The specific repair parts and quantity of each part shall be furnished as specified in the individual equipment specification. If no list is contained in a equipment specification, parts and quantities of parts shall be furnished or considered necessary to maintain the equipment in operating condition during a one year period.

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(c) Where special tools are required for installation and alignment of the generator set, one set of such tools shall be shipped with the first generator set scheduled for installation on any one ship.

4. QUALITY ASSURANCE PROVISIONS

4.1 Resonsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The supplier shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.1 and 6.2) .

4,2 Performance tests.

4.2.1 Unit tests and test agenda. All units shall be individually tested as specified herein. It preferred that these tests be conducted at the place of manufacture prior to final assembly of the generator set. However, if it is more practical to conduct all or part of the required unit tests after final assembly of the generator set, such method of conducting unit tests is acceptable provided all necessary loading and instrumentation facilities are available and tests are conducted in strict compliance with the applicable requirements.

4.2.1.1 Test a ends. A complete test agenda shall be submitted to the ordering activity for review prior to t e start of any tests. This agenda shall list all tests to be conducted on the individual units and assembled generator sets. Each test shall be identified by the applicable specification showing paragraph number and title of test. The agenda shall indicate where each test is to be conducted.

4.2.2 Test reports. Test results shall be recorded on test forms prepared by the manufacturer. Copies of the first unit test reports shall be furnished to the procuring activity in quantities as specified in the contract.

4.3 Electrical tests. Generators, excitation and voltage regulation systems shall be tested in accordance with MIL-G-3124 and MIL-R-2729 and 4.3.1 through 4.3.3 herein.

4.3.1 Electromagnetic interference. Electromagnetic interference tests are not required.

4.3.2 Short circuit. The single phase and three phase short circuit tests shall be conducted with the regulator in the circuit for a period of time as determined by $I_{\lambda}{}^{2}t = 180$, where I_{ℓ} is the sustained value of line current on a per unit basis, and t is time in seconds.

4.3.3 The first set of excitation and voltage regulation equipment components shall be given type I vibration tests of MIL-STD-167.

4.4 First unit tests. Test shall be conducted on the first unit. First unit tests need not be conducted if an engine, identical to the one being offered, has previously been accepted for submarine service and meets the rating requirements specified in 3.2 and has proven to be satisfactory in submarine service. If the engine for which extension of first unit testing is proposed, is not part for part identical to the engine previously tested, the supplier shall submit for review to contracting officer a list of those parts which are not identical. Part number and nomenclature of both old and new parts and reasons for making the change shall be included. The Government reserves the right to require complete retesting after review of the changes. Tests shall be conducted as specified in 4.3.2. No instruments containing mercury shall be used at any time in connection with tests.

4.4.1 Test conditions. The dynamometer load for the engine shall be such as to insure that the engine identification plate output as specified in the contract or order can be

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obtained under standard conditions. The dynamometer load necessary to satisfy this requirement shall be determined using the following formula:

$$BHP_d = \frac{BHP}{\frac{29.92}{P_o}\sqrt{\frac{T_o}{560}}}$$

Where:

BHP_d = Dynamometer load (observed).
BHP = Engine identification plate output.
P_o = Observed ambient pressure in inches of Hg (dry barometer).
T_o = Observed ambient temperature (absolute).

The correction factor shall be applied for all dynamometer test loads.

4.4.1.1 The exhaust and intake system to the engine, shall be provided with valves capable of varying the engine blower intake vacuum from *O* to 7 inches of Hg. and the exhaust pressure up to 15 p.s.i. Means shall be provided to control both intake and exhaust valves manually or by an automatic mechanism which will follow the simulated test cycles specified herein.

4.4.1.2 The temperature cf the raw water to the engine shall be a minimum of 90°F. The engine fresh water shall be treated to prevent corrosion.

4.4.1.3 Fuel used during the tests shall be diesel fuel corresponding to MIL-F-16884.

4.4.2 First unit tests.

4.4.2.1 Surface endurance test. The exhaust back pressure at the engine exhaust outlet and blower inlet vacuum for rateed and speed shall be adjusted to 3 p.s.i. and 1 inch Hg. respectively. The engine shall be run for 125 eight hour cycles as follows (the system shall meet performance requirements and no shutdown shall be permitted due to system malfunction) :

(a) 2 hours at rated load and speed.
(b) 1 hour at 85 percent load and rated speed.
(c) 10 minutes at minimum idling load and speed.
(d) 1 hour and 50 minutes at rated load and speed.
(e) 10 minutes at minimum idling load and speed.
(f) 30 minutes operation at 50 percent load and rated speed.
(g) 10 minutes at minimum idling load and speed.
(h) 10 minutes at 85 percent load and rated speed.
(i) 1 hour and 50 minutes at 110 percent rated load at rated speed.
(j) 10 minutes shutdown (minimum between each cycle).

Engines may be shut-down, as required, between each complete cycle or full cycles may be run on a continuous basis.

4.4.2.2 Fuel consumption test. Fuel consumption tests of 1/4 hour duration with intervals between each run for stabilization of operating conditions, shall be made for each load of O, 40, 80, 100 and 110 percent of rates hp. Fuel consumption shall be corrected for the difference *in* high heat value of the fuel actually used during the test and the standard 19,350 British thermal unit (B.t.u.) per pound.

4.4.2.3 Spectrometric oil analysis. Lubricating oil samples shall be taken from the lube oil scavenge line before oil filter with the engine running. The oil must be circulating and at operating temperature. Samples shall not be taken from a gage line.

4.4.2.3.1 Sampling intervals. Samples shall be drawn during the 2 hours at rated load and speed period of the first and last cycle specified in 4.4.2.1 and during the same period of each ten cycles during the test (total 14 samples).

4.4.2.3.2 A spectrometric analysis of each oil sample taken in accordance with 4.4.2.3.1 shall be made. The concentration of elements in ppm of oil wetted wearing parts shall be recorded. The analysis shall identify and measure the following elements, as applicable, in

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parts per million (ppm) with the degree of accuracy indicated in 4.4.2.3.3.

(a) Iron (Fe).
(b) Lead (Pb).
(c) Copper (Cu).
(d) Chromium (Cr).
(e) Aluminum (Al).
(f) Nickel (Ni) .
(9) Silver (Ag).
(h) Tin (Sri).
(i) Silicon (Si).
(j) Sodium (Na).

4.4.2.3.3 Degree of accuracy. The degree of accuracy shall be as follows:

Standard reference specimen (Range in ppm)	Standard deviation (Maximum in ppm)
3-9	1.5
10-19	2
20-49	3
50-99	5
100-199	8
200-500	15

4.4.2.4 The supplier shall establish maximum limits for exhaust gas temperatures and blower intake and discharge temperature differential. The engine shall then be operated at various snorkel loads and rated rpm at various blower intake vacua and exhaust back pressures in order to determine the engine output limits at which the engine will not exceed exhaust temperature or blower differential temperature limits as established above.

4.4.2.5 Snorkel operation. The engine shall be operated continuously for 160 hours rated snorkel load and rpm. The exhaust back pressure at the manifold outlet shall be 6 psi. Intake air pressure shall be 2 inches Hg. vacuum. Fuel" consumption shall be recorded and submitted to the command or agency concerned.

4.4.2.6 Simulating loss of depth control. The engine shall be operated at the greatest load which will not exceed temperature limits established in 4.4.2.3. Automatic control of the blower intake and exhaust valves shall be arranged to gradually decrease the blower intake pressure from 2 inches of Hg. below atmospheric pressure up to 7 inches of Hg. below atmospheric pressure, and at the same time increase exhaust hack pressure from 6 psi up to 12 psi. The time required to reach these maximum pressure conditions shall be 3.5 minutes. The control valves shall then permit the engine to return to snorkel operations (see 4.4.2.4) in 1 minute. The engine shall operate 5 minutes under snorkel conditions. The entire cycle, therefore, will require 9.5 minutes. The unit shall operate on this cycle 500 times to demonstrate the mechanical reliability *of* the engine.

4.4.2.7 Simulated wave action. The air inlet pressure shall be constant throughout this test and shall be 2 inches of Hg. below atmospheric pressure. The exhaust back pressure shall be regulated for a standard pressure of 6 psi with the damper valve in exhaust line controllable to raise the pressure to 8 psi. The cycle of operation shall be for the pressure to rise to its maximum in a period of 25 seconds, returning to the standard pressure in 5 seconds and a rest period of 30 seconds or a total cycle time of 60 seconds. This cycle of operation shall be repeated for a period of 48 hours at rated snorkel load and rpm.

4.4.2.8 Simulated submerged starting test to remove water from the snokel tube. The engine shall be started with the exhaust line closed until the back pressure reaches a maximum of 20 psi in approximately 6 seconds. The exhaust line shall then be opened to reduce the back pressure to the normal pressure of 6 psi. The engine shall then be shut down, and the test shall be repeated. A total of 250 starts shall be made. One of the starts shall be made from the cold ambient condition after the engine has been shut down for at least 24 hours.

4.4.3 Test data. Test data shall be submitted on supplier's test forms or on large sheets with lines folding to 8 by 10-1/2 inches for filing.

4.4.3.1 All temperature and pressure readings necessary for complete evaluation of the engine performance including ambient room temperature, barometric pressure and relative humidity, shall be entered on the test sheets. Fuel consumption in pounds per hour shall be entered for all runs during which engine conditions to become stabilized. All addition of

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lubricating oil and oil changes shall be recorded. Readings of instruments shall be recorded at intervals not exceeding 1 hour. When a phase of a test is of such short duration that operating temperatures cannot be stabilized, no reading is required.

4.4.3.2 Spectrometric anlysis. The following information/data shall be provided:

(a) List of oil wetted wearing parts and corresponding wear metal elements.
(b) Recommended limits in ppm of the wear elements/metals listed in 4.4.3.2(a). It is the intent to use these values as a basis to segregate normal from abnormal operating machinery.
(c) Tabulation of the analysis data recorded in accordance with 4.4.2.3.2. Tabulation shall include such related and pertinent data as chemical and physical characteristics of the lubricating oil used during the test, initial quantity of oil in the system, hours of operation from start of test at which oil was added\changed and applicable quantity of oil added/changed.

4.4.4 Post trial examination. After completion of all tests, the engine shall be disassembled and examined. For this purpose, all cylinders shall be opened, pistons pulled and the cylinder bores, pistons, piston rings, wrist pins, and crankpin bearings examined for defects and measured for wear. Gear train covers and all crankcase covers shall be removed for examination of internal parts. The engine shall be barred over during examination to determine extent of backlash in gear trains. Mating surfaces of gears and all bearings shall be checked for distress such as pitting and galling. Excessive wear, broken or damaged parts, scored cylinders, burned valves or signs of severe stress or excessive wear shall be cause for rejection of the unit. Excessive wear is defined as wear in excess of the manufacturer's published wear limit for the particular part. Cavitation erosion of cylinder liners or cylinder blocks to the extent that such distress would cause premature replacement of the part shall be cause for rejection of the unit.

4.4.4.1 Special tools furnished with the engines shall be tried out on the engine during the *post* trial examination, before examination can be considered complete.

4.4.5 Complete test report. In addition to the documented results of the post trial examination (see 4.4.4), the complete first unit test report shall be submitted to NAVSEC for review and final acceptance and shall include the following data:

(a) Fuel consumption (see 4.4.2.2).
(b) Spectrometric analysis (see 4.4.3.2).

4.4.6 Duplicate engine routine tests. production engines, duplicates of one which has been tested in accordance with 4.4.2, shall be tested as follows:

(a) Two 8-hour cycles as specified in 4.4.2.1.
(b) 24 hours as specified in 4.4.2.5.
(c) 8 hours as specified in 4.4.2.7.

4.5 Generator set test. The first combined diesel generator set including attached and unattached accessories of each size and design offered for delivery shall be subjected to the lead set tests specified in table II. No lead set tests will be required thereafter as long as identical designs are offered or design changes are minor and do not effect performance. The tests shall be conducted with all the intended electrical equipment and attached and unattached accessory equipment.

4.5.1 All duplicate diesel generator sets shall be given the following set tests as specified in table II.

4.6 Test.

4.6.1 Generator air-gap. The minimum air-gap shall be measured by steel feelers or gages. The measurements shall be made *for* each pole, with at least one of the measurements made at the bottom.

4.6.2 Two hour preliminary run. The generator set shall be operated at full load for 2 hours to bring it up to or near its operating temperature. During this period the general operation of the set shall be observed and any necessary adjustments shall be made.

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Table II - Test for combined diesel generator sets.

Description of tests	Applicable paragraph	Lead set tests	Follow set tests
Generator air gap	4.6.1	X	X
2 hour preliminary run	4.6.2	X	X
5 hour continuous test	4.6.3	X	---
Mechanical balance	4.6.4	X	X
Governor tests	4.6.5	X	X
Range of speed changes	4.6.6	X	X
Brush examination	4.6.7	X	X
Generator voltage (manual control)	4.6.8	X	---
Effectiveness of enclosure	4.6.9	X	---
Torsiograph	4.6.10	X	---
Lubrication	4.6.11	X	X
Starting	4.6.12	X	---
Inclined operation	4.6.13	X	---
Weight and center of gravity	4.6.14	X	---
Mechanical shock test	4.6.15	X	---
Reliability	4.6.16	X	---
Maintainability	4.6.17	X	---
Parallel operation	4.6.18	X	---

4.6.3 Five-hour continuous test at full load. Immediately following the test specified in 4.6.2 the-r set shall be operated at full load for 4 hours. The following data shall be taken at 30 minute intervals during the test and shall be recorded on test record forms:

(a) Time.
(b) Ambient temperature.
(c) Fuel consumption.
(d) Engine fuel oil pressure.
(e) Engine lubricating oil pressure.
(f) Engine lubricating oil temperature.
(g) Water temperature.
(h) Speed of engine.
(i) Generator bearing temperatures.
(j) volts , line-to-line.
(k) Current, line.
(1) Generator frame temperature.
(m) Generator armature temperature.
(n) Generator field current and voltage.
(o) Generator inlet and exhaust air temperature.
(p) Frequency.
(q) Kva and kw.
(r) Power factor.

4.6.3.1 At the conclusion of the 5 hour period (see 4.1.3), temperatures of the rotating components of the generator shall be measured and shall be recorded immediately upon shutdown.

4.6.4 Mechanical balance. The mechanical balance shall be observed during the progress of all test. Any unusual noises, excessive vibration, or other indication of malfunction shall be cause for rejection of the unit. Vibration readings in roils double amplitude should be taken on the engine and generator bearing brackets or shafts, and recorded for information.

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4 .6.5 Governor tests.

4.6.5.1 Steady state speed regulation. Generator sets shall be tested as follows for steady state speed regulation:

(a) At no load, and with manual or automatic control of generator excitation, the generator set shall be adjusted to rated speed.
(b) The load shall be changed in 10.O percent increments from no load to 100 percent load and back to no load. At each point, the voltage shall be adjusted to rated value and the speed shall be recorded.
(c) The speed regulation curve shall be plotted and furnished as part of the test data.

4.6.5.2 Steady state governing speedband. At O, 25, 50, 75 and 100 percent rated load, the steady state governing speedband shall be observed and recorded.

4.6.5.3 Momentary underspeed and overspeed. Momentary underspeed and overspeed tests shall be conducted as follows:

(a) The generator set shall be adjusted to rated speed, voltage, and load. Rated load shall then be removed in one step.
(b) The generator set shall be operated at rated speed, rated voltage, and no load. A 100 percent resistance load shall then be suddenly applied in one step.
(c) Readings of overspeed, underspeed, arid recovery time shall be recorded. The method of measuring the speed changes and recovery time shall be approved by the command or agency concerned. During these tests, manual or automatic . control of generator excitation may be used. If manual control is used, the no-load voltage for the underspend test shall be set at 105 percent of rated voltage.

4.6.5.4 Overspeed trip. During the overspeed trip test, the regulating governor shall be inoperative. The engine speed shall be increased above rated speed until the overspeed trip acts to determine the r.p.m. at which the engine shall be shutdown.

4.6.6 Range of speed changer. Range of speed changer test shall be conducted as follows:

(a) At no load and with manual control of generator excitation, the generator set shall be adjusted to rated speed and voltage. Operate the speed changer so that the speed is at the minimum specified value. Record speed. Operate the speed changer so that the speed is at the maximum specified value. Record speed and rate of change from minimum to maximum speed.
(b) At 100 percent load and with manual control of generator excitation, the generator set shall be adjusted to rated speed and voltage. Operate the speed changer so that the speed is at the minimum specified value. Record speed. Operate the speed changer so that the speed is at the maximum specified value. Record speed and rate of change. The load and voltage may be maintained at rated valves when measuring maximum speed.

4.6.7 Brush examination. During all tests, the performance of brushes should be noted at each load point. Particularly, it should be noted if any injurious sparking occurs when the entire load is dropped or applied in one step.

4.6.8 Generator voltage under manual control. Tests shall be conducted as follows with the general field cold. the voltage regulator shall be made inoperative by positioning the manual-automatic voltage regulation switch in the manual position.

(a) Operate the unit at a speed corresponding to no load. Adjust the manual control so that the voltage is at the minimum specified value. Record voltage and speed. Adjust the manual control so that the voltage is at the maximum specified value. Record voltage and speed.
(b) Operate the unit at no load, and adjust the voltage to the minimum specified valve. Add load (at rated p.f.) and adjust the voltage, Speed, and load until the unit is operating at the minimum specified voltage with rated load. Record voltage, current, and speed.
(c) Reduce the voltage and load to rated voltage and no load. Adjust the voltage to the maximum specified value. Add load (at rated p.f.) and adjust voltage, speed, and load until the unit is operating at the maximum specified voltage with rated load. Record voltage, current, and speed.

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4.6.9 Diffectiveness of enclosure. Enclosure tests shall be conducted in accordance with MIL-E-2036 on the electrical portion of the generator set, where it is evident that the effectiveness of any enclosure-has been altered by assembly as a part of the generator set.

4.6.10 Torsiograph test. Generator set shall be tested in accordance with type III requirements of MIL-STD-167.

4.6.11 Lubrication. Performance of the lubricating system shall be determined by a careful observation and general examination of the parts. Limitation of the generator bearing temperatures, as specified in MIL-G-3124 shall not be exceeded. This test may be made at no-load provided the bearing load is not influenced appreciably by the load condition of the generator. Lubricant similar to that required for service operation shall be used.

4.6.12 Starting tests.

4.6.12.1 Cold start. Prior to conducting the cold start test the diesel generator set shall have been shutdown for at least 12 hours. Cold water, at a temperature not exceeding 35°F., shall be circulated through the engine for not less than 1 hour or until the outlet temperature of the water does not exceed 38°F. Compliance with the requirements of 3.1.14 shall be demonstrated.

4.6.12.2 Normal starting. Immediately following the cold start test (see 4.6.12.1) nine conseculative starts shall be made under normal ambient conditions. The average time required to start the unit, and the minimum starting r.p.m. shall be determined. The engine shall fire and continue operating within 5 seconds after the starting mechanism is engaged.

4.6.12.3 The following data shall be obtained:

(a) Minimum quantity of air expressed in cubic feet of free air at 14.7 p.s.i., and 68°F., for the cold start.
(b) Minimum quantity of air expressed in cubic feet of free air at 14.7 p.s.i., and 68°F., necessary for 10 successive starts, including the cold start.
(c) Minimum air pressure at which the unit will start within the 5 second time limit.

4.6.13 Inclined operation. The inclination test shall be made with the unit running at 60 cycle speed with or without load for at least 30 minutes in each of the following positions:

(a) Shaft inclined, forward end down 7 degrees.
(b) Shaft inclined, forward end up 7 degrees.
(c) Shaft horizontal, base inclined to right 15 degrees.
(d) Shaft horizontal, base inclined to left 15 degrees.

4.6.13.1 During the tests of 4.6.13, the lubricating oil sump(s) shall be filled to the maximum operating level. It shall be ascertained that the mechanical balance is as good as it was in the horizontal position, that there is no pounding or grinding of the bearings, and that there is no leakage of oil or cooling water.

4.6.14 Weight and center of gravity. The generator set and all detached accessories shall be weighed dry. The center of gravity for the generator set (if feasible) shall be determined by actual measurement. If-measurement of the complete generator set is not feasible, its center of gravity shall be calculated using weights and center of gravity figures for components obtained by actual measurement.

4.6.15 Mechanical shock (see 6.1). The complete diesel generator set shall be tested in accordance with the requirements of the Appendix to this specification.

4.6.17 Maintainability demonstration. The reliability demonstration test, if required, shall be as specified (see 6.1).

4.6.17 Maintainability demonstration test. A maintainability demonstration test shall be conducted to verify the achievement of the quantitative value of predicted maintainability specified in 3.1.2.7. The demonstration shall be conducted in accordance with MIL-STD-471 (see 6.2) except as modified herein.

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4.6.17.1 A demonstration plan shall be submitted by the manufacturer to the purchaser for review and acceptance. This plan shall be formulated so that it will apply to a basic design or designs and any known modifications thereto. Once this plan has been accepted, it shall be used on generator sets furnished under this specification except where a new basic design *or* modification to the original basic designs have been incorporated in generator sets and are not covered by the original plan. In this case, the generator set manufacturers shall submit a plan for the new basic design or the modification, as applicable, to the purchaser for acceptance. As a minimum, the following items shall be covered:

(a) Preventative maintenance tasks.
(b) Corrective maintenance tasks which would normally be accomplished onboard ship, such as replacement of bearings and seals, water pump seals, cooler zincs, governor parts, safety devices, gages, thermometers, brushes and springs, heaters, etc.

4.6.17.2 *Test* conditions. The maintainability test shall be performed with the equipment or system Installed In a manner which simulates an actual shipboard installation and the access envelope (see 3.1.1.2.1) shall be simulated with panels and screens. Test equipment tools, repair parts and maintenance literature, at least in the form of preliminary manuals covering maintenance instruction shall be made available for the tests. Test team, facilities, and support material shall represent the normal shipboard resources. Test mechanics shall be *given* no outside assistance unless approved by the purchaser. The following shall be recorded, monitored, and certified by the purchaser's representative:

(a) Data collected, including man-hours, clock-hours, maximum number of active mechanics, part identification, and documentation of the specified maintenance tasks.
(b) Factors which influenced the data.
(c) Identification of other parts removed or moved to permit replacement of the part which a replacement time has to be demonstrated.
(d) Computation or measurement of the required replacement times
(e) Deficiencies.
(f) Recommendations.
(g) Results of retest, if applicable.
(h) Certification by a purchaser's representative of the data obtained and whether or not the maintenance task time requirements have been met.

4.6.17.3 The training and skill levels of the manufacturer's shop personnel who perform the maintenance tasks shall be commensurate with Navy personnel who will maintain the generator set.

4.6.17.4 Simulation of faults or failures by the introduction of defective parts is not required.

4.6.17.5 The purchaser and NAVSEC shall be notified at least 30 days in advance of the scheduled test so that arrangements can be made to witness the test if desired.

4.6.18 Parallel operation test. Parallel operation tests shall be conducted to the extent necessary to demonstrate stale transfer of real and reactive power from one set to the other and with the voltage plus or minus 5 volts of rated.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3.)

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Level A.

5.1.1 Generator and voltage regulators. Generators and voltage regulators shall be cleaned and preserved level A in accordante with MIL-E-16298, MIL-R-2729 and MIL-E-17555, as applicable.

5.1.1.1 Generator and voltage regulators. The basic diesel engine shall be cleaned and preserved in accordance with IV, method II, of MIL-E-10062 to the extent specified. Manifolds, pumps, turbochargers and other attached accessories shall not be removed from the engine during cleaning and preservation unless the removal is considered essential for complete coverage of the preservative.

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5.1.1.2.1 Lubricating oil, fresh and sea water systems. Preservative type I, grade 30, in accordance with MIL-L-2160, shall be used throughout thr engine and gear, fresh water, sea water, and lubricating oil systems in lieu of the P-2, P-3, and P-9 preservative specified in MIL-E-10062. When required, engine manufacturers may supplement MIL-L-21269 with a maximum zinc content limit. Clean filter elements shall be installed in the fuel and lubricating oil systems after preservation is completed. In addition, the fresh water system with thermostats removed, shall be drained and completely dried with warm air prior to preservation. If it is impractical to perform a drying-out procedure, the cooling system shall be flushed with a soluble oil conforming to MIL-I-24453, which will emulsify the water and remove it on draining. The system shall then be flushed with the specified preservative and the thermostats replaced.

5.1.1.2.2 Fuel oil system. Preservative MIL-L-21260, type I, grade 10, shall be used throughout the fuel system.

5.1.1.2.3 Valve mechanism. Access covers shall be removed and all surfaces within the valve compartment, including rocker mechanisms, valve stems, springs, guides, push rods, and the inside face of the cover plate shall be coated with P-10 preservative.

5.1.1.3 Packaging. The diesel generating set with attached accessories shall be individually unit protected in accordance with MIL-P-116, method II. As specified (see 6.1), one of the following two packaging methods shall be used:

(a) Method II a - nonreusable floating bag. The flexible barrier bag shall conform to the requirements of MIL-P-116 and MIL-B-131,class 1. Humidity indicators shall conform to MIL-I-26860, (plug type). All projections and sharp corners of the engine unit shall be cushioned as specified in 5.2.3. Whereever hold-down bolts go through the barrier bag, gaskets fabricated of material conforming to G-1123 of MIL-G-12803 shall be secured to both the inside and outside of the bag with adhesive conforming to MMM-A-260 for fiber gaskets and MMM-A-1617 for rubber gaskets. Desiccant shall be uniformly distributed throughout the package to obtain optimum effectiveness.

(b) Method II a reusable floating bag. A reusable container conforming to MIL-C-9959, type III, shall be used.

5.1.1.4 Detached equipments. Detached voltage regulator and excitation components, accessories, instruments and special tools shall be preserved-packaged level A in accordance with the commodity specification, MIL-R-196, MIL-R-2729 or MIL-E-17555, as applicable. Unless used in sets, pairs, or quantity greater than one or as specified in the contract or order, detached instruments, accessories, repair parts and tools shall be individually preserved-packaged.

5.1.1.5 Repair parts (onboard and system stock). The level A preservation-packaging methods shall be in accordance with MIL-STD-758.

5.1.2 Level C. Preservation-packaging of generator sets, detached components, accessories, instruments and tools shall afford protection against corrosion? deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A or C, as specified (see 6.1).

5.2.1 Basic generator set.

5.2.1.1 Level A. Each unit shall be packed in a reusable plywood sheathed crate conforming to type II, class 2, style a or b of MIL-C-104. Each unit shall be anchored, blocked and braced within the shipping container to resist shock and prevent damage during shipment and storage. Anchoring, blocking, and bracing shall be in accordance with 5.2.3 herein and the appendix to MIL-C-104. Crates shall be provided with an inspection port for use in viewing the humidity indicator mounted in the barrier material. The rubbing strip of MIL-C-104 shall be three inch in lieu of two inch thick material.

5.2.1.2 Level C. Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers *or* method of packing shall conform to the Uniform Freight *or* National Freight Classification Rules or other carrier regulations, as applicable to the mode of transportation.

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5.2.2 Detached components, accessories, repair parts instruments and tools. Detached components, accessories, repair parts instruments and tools shall be packed level A or C, as specified (see 6.1) in accordance with the applicable commodity specification, MIL-R-2729 or MIL-E-17555, as applicable. Rough- handling test as-specified in MIL-E-17555 is not required. Repair parts and tools shall be separately packed and shipped concurrently with the basic equipment, set or system. Detached instruments, components and accessories shall be packed separately and secured in the engine shipping crate if space permits or shipped concurrently with the basic equipment.

5.2.2.1 Index list. An index list shall be inserted in each shipping container containing repair parts or tools accompanying the equipment or procured as a set or kit. The list shall be placed in a transparent waterproof bag and heat scale?. The list shall be suitably placed in the inside of the shipping container for quick accessibility. The list shall completely itemize the contents of the container.

5.2.3 Anchoring, blocing, bracing and cushing. Anchoring, blocking, bracing and cushioning of engine units, detached, components, accessories, and tools shall be in accordance with MIL-STD-1186 and the applicable commodity or container specification and appendix thereto. In addition, the following is mandatory for cushioning materials when used:

(a) Use of loose excelsior or shredded newspaper is prohibited unless such are enclosed in a sealed, waterproof barrier conforming to PPP-B-1055 or MIL-B-13239.

(b) Polystyrene “loose-fill” materials for level A packing applications such as cushioning, filler, dunnage, etc., is prohibited. Unless acceptable to the procuring activity (see 6.1), use of polystyrene “loose-fill” for packaging and packing under 5.1.2, 5.2.1.2, and 5.2.2 (level C) for applications such as cushioning, filler, dunnage, etc. , is prohibited. When acceptable for use under 5.1.2, 5.2.1.2, and 5.2.2, (level C) packages and containers (interior and exterior) shall be marked or labeled as follows:

“CAUTION

Contents cushioned etc., with polystyrene (loose-fill) material.
Not *to* be taken onboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber,
fiberboard, or transparent flexible cellular material. ”

5.3 Marking,

5.3.1 Standard markings. In addition to any special marking or instructions required herein or by the contract or order (see 6.1), unit and intermediate packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Special markings.

5.3.2.1 Method II. Method II packages shall be marked in accordance with MIL-STD-129.

5.3.2.2 Shipping containers. Shipping containers shall have the following markings:

(a) Adjacent to method II markings:

“STORE RIGHT SIDE UP - WARNING - SEE UNPACKING INSTRUCTIONS.”

(b) Apply adjacent to the identification marking on the side of the container:

“CAUTION - THIS EQUIPMENT MAY BE SERIOUSLY DAMAGED UNLESS UNPACKING INSTRUCTIONS ARE CAREFULLY FOLLOWED. UNPACKING INSTRUCTIONS ARE LOCATED (state where located).”

(c) “REUSABLE INTERIOR AND EXTERIOR CONTAINER (when applicable) .“

(d) Handling and structural markings as applicable (see MIL-STD-129 and appendix to MIL-C-104).

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5.4 Special instructions.

5.4.1 Preserved units. All generator sets preserved level A shall be tagged as follows:

"The fluid systems of this generator set have been preserved with type P-10 preservative. No special depreservation procedures are required with the exception of the engine fresh water system. Depreservation of the fresh water system shall be in accordance with chapter 9412, paragraph 9412.118 of NAVSHIPS 0901-412-0002."

5.4.2 Depreservation. The equipment supplier shall provide any additional depreservation instructions for systems or accessories when such instructions would not be applicable to 5.4.1 or would be necessary due to design peculiarities of his equipment. Instructions shall be packaged in a transparent waterproof plastic bag, minimum 4 mil thick. Closure shall be by heat sealing. The shipping container in which the instructions are packed shall be marked to so indicate.

5.4.3 Unpacking. Unpacking instructions shall be provided for complex equipment or system and floating bag type packs, the instructions shall contain the following information:

"To unpack remove the top and sides, leaving the unit resting on the bottom of the packing case. Remove the packing case and slip the unit off the base. In unpacking the item, the following precautions shall be observed to prevent possible damage:

(a) Observe the arrows marked on the shipping container. These point to the cover which can be removed most readily.
(b) Remove nails with a nail-puller only.
(c) Remove screws with a screwdriver only.
(d) Never pound *or* hammer the shipping container.
(e) Keep all levers and crowbars away from the interior of the container."

5.4.3.1 These instructions shall be placed in a sealed waterproof envelope prominently marked "UNPACKING INSTRUCTIONS" and firmly affixed to the outside of the shipping container in a protected location, preferably adjacent to the identification marking. If the instructions cover a set of equipment packed in multiple containers, the instructions shall be affixed to the number one container of the set or system.

5.4.4 Technical manuals. Depending on the type of packaging specified for the generator set unit, the accompany technical manuals shall be packaged as specified in 5.4.4.1 through 5.4.4.3.

5.4.4.1 Nonreusable bag . Manuals shall be packaged in transparent plastic bags minimum 4 mil thick and heat scaled . Manuals shall not be placed within the sealed MIL-B-131 bag used to encase the generator set but placed on the inside of the plywood crate. Packing list shall indicate the approximate location *of* the manuals. The container shall be marked "Manuals Enclosed". For ease of removability, the location of the manual(s) shall be such that they are readily accessible when the container is opened.

5.4.4.2 Reusable bag. Technical manuals shall be placed in the pocket provided on the outside of the resuable bag. The pocket(s) shall be marked "Manual Enclosed".

5.4.4.3 Bulk shipment. Technical manuals shipped in bulk quantities shall be packaged and packed in accordance with the requirements of MIL-M-15071.

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Rated snorkel and surface loads operation (see 3.1.5).
(c) Speed required (see 3.1.5).
(d) Inlet vacuum and exhaust back pressures, if specified (see 3.1.5.1).
(e) Base load required (see 3.1.5.1 (a)).
(f) Weight and dimensional construction required (see 3.1.8).
(g) Pressure required for ship air supply (see 3.2.5.7).
(h) Submergence pressure required (see 3.2.5.9(a)(2)).
(i) Kw specified (see 3.3).
(j) Submergence design and test procedures (see 3.3.3).
(k) Mounting for switchboard enclosures (see 3.4.2).

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(l) Preliminary drawing required (see 3.5.4) .
(m) Type and quantity required for microfilm (see 3.5.5).
(n) Tabulating index cards required (see 3.5.5.1).
(o) Repair parts and special tools required (see 3.7).
(p) If reliability demonstration test is required (see 4.6.16).
(q) Level of preservation-packaging specified (see 5.1).
(r) Packaging methods specified (see 5.1.1.3).
(s) Level of packing specified (see 5.2).
(t) Whether use of polystyrene "loose-fill" is acceptable (see 5.2.3).
(u) Special marking required (see 5.3.1).

6.2 Management control system documents. The following management control system documents should be included on DD Form 1660:

(a) MIL-STD-470 (see 3.1.2.1).
(b) MIL-STD-785 (see 3.1.2.1).
(c) MIL-I-45208 (see 4.1.1).
(d) MIL-STD-471 (see 4.6.17).

Preparing activity:
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(Project 6115-N311)

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NOTES

1- FOR TWIN ENGINE PRIMEMOVERS, A STRAINER ON THE SUCTION SIDE OF THE PUMP MAY BE SUBSTITUTED FOR THE SIMPLEX METAL-EDGE STRAINER PROVIDED THE FILTER IS RELOCATAED TO THE OUTLET SIDE OF THE HEAT EXCHANGER.

2- THE SIMPLEX METAL EDGE STRAINER SHALL BE DESIGNED IN A MANNER TO PREVENT THE ACCUMULATED CONTAMINENTS IN THE STRAINER SUMP FROM ENTERING THE LUBE OIL SYSTEM THROUGH THE RELIEF VALVE. AN INTERNAL STRAINER RELIEF VALVE CAN BE FURNISHED IN LIEU OF AN EXTERNAL RELIEF VALVE IF APPROVED BY THE PURCHASING ACTIVITY.

3- THE FILTER EXTERNAL RELIEF VALVE SHALL BE SET TO OPEN AT NO MORE THAN 25 PSI. DIFFERENTIAL PRESSURE AND TO CLOSE AT NO LESS THAN 20PSI. DIFFERENTIAL PRESSURE. THE FILTER BY-PASS SHALL HAVE THE ABILITY TO CARRY 100% OF THE PUMP CAPACITY AT A DIFFERENTIAL PRESSURE OF NO MORE THAN 30 PSI. ACROSS THE SYSTEM.

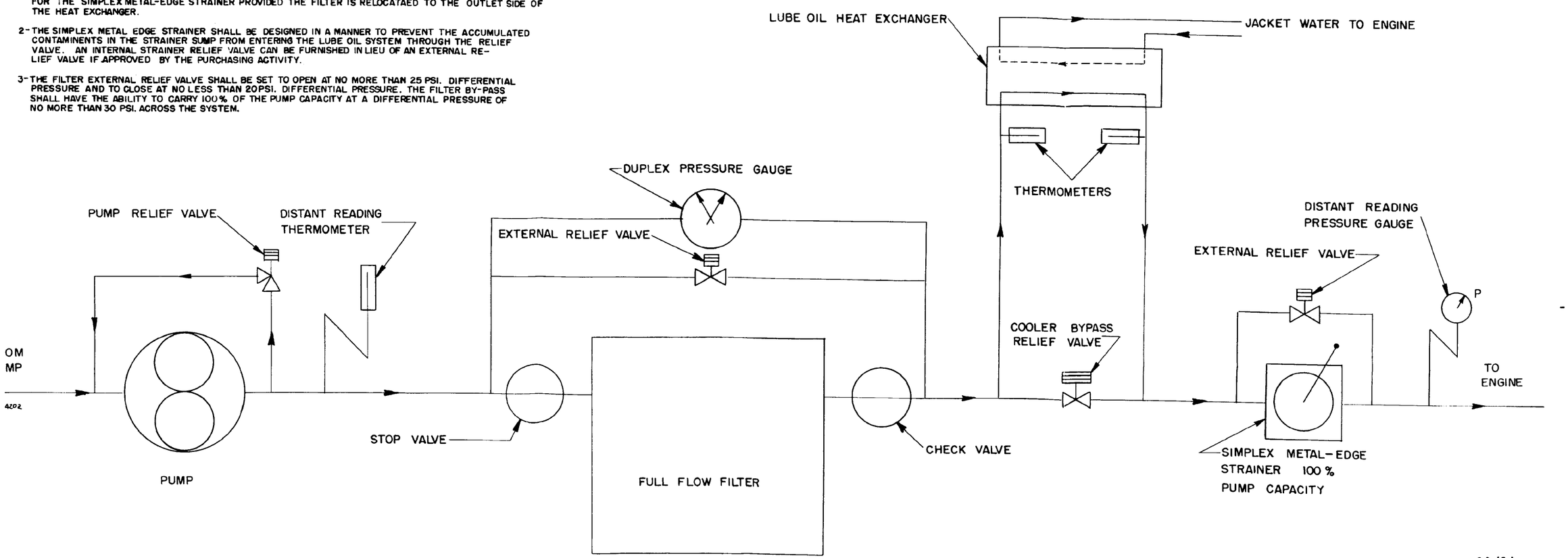


Figure 1 - Full flow lube oil filter system.

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APPENDIX

REQUIREMENTS FOR HIGH IMPACT (HI) SHOCK EQUIPMENT

10. SCOPE

10.1 This appendix covers the design, test, and acceptance requirements for diesel generator sets which must withstand HI shock.

20. APPLICABLE DOCUMENTS

20.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the appendix to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, HI (High Impact) Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-S-21952 - Steel (HY-80 and HY-100) Bars, Alloy, Hot Rolled.
MIL-S-22664 - Steel, Alloy, Extrusion, Structural Shapes (HY-90 and HY-100).
MIL-S-22958 - Steel, Alloy, Shapes, Hot Rolled Structural (HY-90 and HY-100).
MIL-S-23008 - Steel Castings, Alloy, High Yield Strength (HY-80 and HY-100).
MIL-S-23009 - Steel Forgings, Alloy, High Yield Strength (HY-80) and HY-100).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 The requirements of MIL-S-991 shall apply except as otherwise specified herein.

30.2 Diesel generator sets and all attached and detached accessories shall conform to the requirements for type A, class I of MIL-S-901 and the shock test criteria specified in 30.4. Acceptance of generator set and accessory designs which have been subjected to shock tests shall be based on the ability of the equipment to conform to acceptability criteria, as revealed by its performance during and after shock tests and the results of post-test examination and operating tests conducted at the supplier's plant. In the event that the acceptance criteria are not met, acceptance will also depend upon the corrective action proposed by the contractor. A detailed report of the inspection along with the contractor's proposals for shock hardening (if necessary) shall be forwarded to NAVSEC. Review by NAVSEC of the results of shock tests, the pre-shock and post shock test inspections, and the corrective action shall be required.

30.3 Intended function. The intended function of shockproof generator sets is to continuously deliver electrical power conforming to required performance characteristics throughout and after being subjected to shock of the magnitude experienced during shock tests as specified in MIL-S-901. Further, a shockproof generator set, when subjected to shock, shall not suffer damage to the extent that it creates a possible hazardous situation such as fire or injury to operating personnel, or such as to result in likely failure of major units of the set if the set *if* not immediately shut down for corrective action. In determining if equipment has passed shock tests, the criteria in 30.4 herein shall be used.

30.4 Acceptability criteria for shock tested equipment. In determining if a shock tested equipment has passed shock tests, the criteria specified hereinafter shall apply. NAVSEC shall be the final judge of whether an equipment shall be accepted or rejected on the basis of shock test requirements: however, the manufacturer shall have the option of convening a conference with NAVSEC to discuss questionable situations involving acceptance or rejection of shock-tested equipment.

30.4.1 Damage as characterized or as specified herein shall not be permitted as a result of shock tests. In the following, as elsewhere in this section, shock tests are sometimes equated to an attack upon a Navy ship.

(a) Damage which causes or necessitates reduction of equipment output to a level below rated for any length of time within 30 days after attack.

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(b) Damage that renders inoperative any item or device installed to protect the equipment or operating personnel, such as speed limiting governors, overspeed trips, low rpm shutdown devices and high temperature low lube oil pressure alarms.
(c) Shock-induced actuation of safety devices specified in 30.4.1, item (b).
(d) Damage or shock-induces misadjustment of speed governing systems that results in inability to maintain speed automatically within the limits imposed by the individual equipment specifications until repairs or adjustments are made.
(e) Loosening or shifting of any item affixed to a rotating part.
(f) Damage that causes any assembly or subassembly to come adrift from any of its intended restraints.
(g) Damage that requires repairs of any type to be made before equipment operating parameters can be adjusted to suit operational requirements.
(h) Damage that creates a safety hazard to personnel or equipment.
(i) Complete fracture of any unit or welded assembly.
(j) Complete stripping or complete failure by gross distortion of all engaged threads (male or female) of any threaded part.
(k) Increase of dynamic unbalance of any rotating part or assembly to such an extent that the vendor must re-balance the affected item prior to delivery of the equipment or prior to conducting field balancing operations.

30.4.2 Damage from shock tests as characterized or specified herein will be permitted to the extent or under the conditions listed for each item.

30.4.2.1 Damage that shortens the useful life of wearing parts by initiating or aggravating some long-term detrimental effect such as wear, fatigue, or erosion is permissible if the useful life of wearing parts is, reduced by no more than 25 percent.[1/]

30.4.2.2 Damage resulting in minor leaks of fluids or gases is permissible insofar as no portion of this appendix is violated by any consequences of the leakage.

30.4.2.3 Loosening of hold-down is permissible if no portion of this appendix is violated as a consequence of such loosening.

30.4.2.4 Damage that brings about a requirement to adjust control apparatus or equipment setpoints in order to regain the pre-attach mode of operation, or damage that brings about a requirement for timely repairs or compensatory action in order to prevent further (and eventually unacceptable) damage that is related to increased wear rates or other long term effects, is permissible only if the following conditions are met:

(a) If NAVSEC agrees that the damage or its effects are of such a nature that timely and proper damage diagnosis and compensation would certainly be made by ship's force personnel familiary with the equipment. (For purposes of this appendix, it is considered that misalignment (if any) or shafting cannot be corrected by ship's force.)
(b) If the equipment status before adjustment/repairs are made is such that no portion of this appendix is violated by the immediate effects of the damage or by requirements for repair of the damage.

30.4.2.5 Damage that results in continuous or intermittent rubbing contact between stationary and moving parts (during equipment operation) in areas where it was not originally intended that such contact be made is generally not permitted. NAVSEC may choose to permit on a case basis, rubs which quickly wear away without causing adverse effects that otherwise violate any portion of this appendix.

[1/] Due to difficulties associated with estimating reduction of useful life, equipment acceptability shall be determined with consideration for the accuracy of the estimate. In cases where it is especially difficult to estimate percent reduction of useful life, equipment will not be rejected for this cause unless in the opinion of NAVSEC that the limits specified herein have been grossly exceeded. In cases where the availability of test or maintenance information permits a more accurate estimate of useful life reduction to be made, appropriately less leniency will be allowed when determining acceptability of the equipment *of* this basis.

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30.4.2.6 Damage that brings about (until repairs are made) a requirement for slight to moderate increase *of* manual effort, maintenance, or attention on the part of equipment operators in order to maintain the operating mode or the machinery space environment may be permissible, when specifically agreed to by NAVSEC.

30.5 Design.

30.5.1 The generator set hold down bolt holes provided for securing equipment to the ship's foundation shall be designed for use with nonfitted bolts on one side and fitted bolts on the other. Maximum clearance between bolt holes and bolts shall not exceed the following:

Nominal bolt diameter	Maximum diameter of hole
3/4 inch or smaller	Nominal bolt diameter plus 1/37 inch
Larger than 3/4 inch	Nominal bolt diameter plus 1/16 inch

30.5.2 When high yield strength materials (materials with yield strength of 75,009 p.s.i. or more) are used, by reason of choice or requirement, to meet design stress levels, or to satisfy resistance to shock, or both, HY-80, or HY-100 in, accordance with MIL-S-21952, MIL-S-22664, MIL-S-22958, MIL-S-23008 or MIL-S-23009, as applicable, shall be used.

40. QUALITY ASSURANCE PROVISIONS

40.1 General inspection. Complete generator sets and detached accessories shall be subjected *to* and successfully pass shock tests.

40.2 Test requirements.

40.2.1 All shock tests shall be conducted in accordance with the grade A machinery requirements and type A tests of MIL-S-901 and as follows:

(a) The generator set package (prime mover, generator, and all attached accessories) and any detached accessories needed to control and operate the set shall be shock tested as a principal unit if the weight size and arrangement limitations are not exceeded. Detached accessories not tested with the generator set package shall be individually tested using the applicable shock testing machine referenced in MIL-S-901 and in accordance with the individual equipment specification (for example, MIL-R-2729 - Regulator Sets, Voltage).

(b) Assemblies or sub-assemblies comprising the equipment to be shock tested shall be those actually made for and furnished under the present contract.

(c) All assemblies and subassemblies comprising the equipment furnished for shock tests shall comply with individual equipment specification requirements. All required performance tests shall have been satisfactorily completed prior to shock testing. 2/

(d) The manufacturer shall carefully inspect equipment which will be shock tested in order to detect any flaws or undesirable conditions that might contribute to shock test damage, 3/ or be mistaken for shock damage during the post-test inspection.

(e) All running clearances, alignments, runouts, and mechanism set points 4/ that are normally established within the equipment and its ancillary devices by the contractor shall be properly established in the equipment before shock tests, even though it may be the contractor's usual practice to establish certain of these settings after installation of the equipment onboard ship. If the contractor desires, setpoints may be established by setting or adjusting the device in question in such a manner that the correct setpoints are theoretically achieved (that is achieved without conducting actual tests).

2/ Request for exceptions to 40.2.1(c) requirements will be considered on a case basis by NAVSEC.

3/ For purposes of this appendix "damage" is defined as any permanent change of the status or condition of any equipment that results from shock testing.

4/ "Setpoints" define the input-output relationships of any mechanisms or device associated with the equipment to be shock tested. For instance, the setpoint or an overspeed tripping device could be "actuation (output) at 19,700 r.p.m. (inPut). "

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If setpoints are obtained in this manner, however, the contractor shall document the pre-test status of all adjustments, dimensions, and characteristics that could possibly cause change of setpoints as a result of shock test damage. If the contractor desires that certain aliqnments, clearances, or setpoints be established by the shock testing facility as the equipment is erected for shock tests (necessary, for instance, if equipment is disassembled for shipment) , the cognizant shock testing facility shall be so informed at least 2 months before the equipment is to be delivered to the shock testing facility. Due to facility limitations, the contractor may be required to provide personnel or equipment, or both, necessary to accomplish such work.

40.3 Equipment mounting during test

40.1 Heavy weight equipment. Shock tests shall be conducted on the floating shock test platform specified in MIL-S-901.

40.3.1.1 Where equipment is expected to be used in various shipboard applications involving different locations, the generator set package shall be mounted on a typical ships foundation welded to the inner dock of the test barge. Equipment intended for one particular application only may be mounted on a foundation simulating that involved in the installation. Deck mounted detached accessories, if tested with the generator set package shall be mounted on a fixture designed to accommodate the accessory and welded to the inner deck.

40.4 Test procedures. Shock tests shall be conducted in accordance with MIL-S-901 and as specified hereinafter.

40.4.1 Equipment need not be operated or energized durinq each blow or shot except for the last blow or shot required by MIL-S-901. Prior to each blow or shot, however, the generator set shall be run at rated speed and voltage, no load, for a period of time necessary to determine that all components are functioning normally. In addition the generator set shall be operated at rated speed and voltage, no load during the last blow or shot. components and accessories, attached or detached, which contain a fluid during normal operation, (such as coolers, and so forth) shall contain a fluid of comparable weight during all blows. The equipment shall be examined for damage after each blow. Correction of minor damage during the progress of tests will be permitted and the test may proceed to the next level of shock after the minor damage is corrected. Major damage shall be cause for rejection and suspension of tests until such time as the suppliers proposed corrective action is approved by the command or agency concerned. In such cases, after corrective action is completed, re-test will be required starting at the test level which caused the damage. The decision as to whether damage is minor or major shall be made by the representative(s) of the command or agency concerned at the test site. In case of disaqreement, the matter shall be referred to the command or agency concerned for final decision.

40.4.2 Instrumentation and measurements durinq the tests shall be as follows:

(a) Velocity pickups, high speed movies, instrumentation for monitoring set operation, and provision for remote shutdown shall be installed.
(b) Initial start-up and check-out shall be conducted under guidance of the manufacturer .
(c) During the last blow or shot an oscillograph recording shall be made showing generator output voltage and frequency characteristics from a time beginning with 5 seconds before the blow until 30 seconds after the blow.
(d) Scribe lines or punch marks shall be used to indicate exact location of all major attached accessories and components (such as blowers, pumps, generator feet, end covers and so forth) prior to shock tests so that any relative motion during shock blows can be measured and recorded.
(e) Generator shaft run-out shall be measured before starting test and after each blow.
(f) Air-gap measurements shall be taken before start of tests and after each blow.
(g) Crankshaft deflection shall be checked prior to start of test and after each blow , if feasible (measurement to be made by the vendor's representative).
(h) With the set operatinq at rated speed, first order vibration levels shall be measured with a vibration pick-up in the X, Y and Z directions before and after each shot.
(i) Insulation resistance and dielectric strength shall be measured prior to start of testing and after final blow.
(j) A thorough visual examination shall be made after each blow and any indications of movement or distortion recorded.

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(k) *Generator cooler covers* shall be removed after each shot for Visual examination of the cooler and any excitation parts.
(l) provision shall be made for remote observation of lubricating oil pressure to bearings and for remote shutdown of the equipment during the last shot.
(m) Additional instrumentation shall be provided by the testing activity as required.

40.5 Post shock test examination and reports.

40.5.1 After completion of shock tests, the equipment shall be returned to the suppliers plant for a complete teardown and examination for damage. Upon receipt in the suppliers plant, prior to removal from the carrier, the equipment shall be given a visual examination to assure that no damage has occurred during transit. The equipment shall then be removed to the test floor area and all examination covers of all components removed and the components shall be examined to determine tunability of the set. If the examination reveals no damage which would prohibit full power operation, the set shall be installed on a test stand and operated at full load for a minimum of 4 hours, Engine and electrical equipment performance shall be monitored and recorded during the load run. All safety devices, such as overspeed trip, low r.p.m. trip, etc., shall be tested in an appropriate manner three times to determine if they are operating properly at the correct set point(s). Upon completion of the tests, the set components shall be disassembled part by part, making visual examination, dimensional checks of parts and clearances as necessary? recording damager evidence of shifting, rubbing, momentary contacts and so forth. Piping and coolers shall be given hydrostatic tests for evidence of any leaks.

40.5.2 A detailed report of the examination and tests shall be prepared and 5 copies shall be submitted to the command or agency concerned for acceptance. The report shall include:

(a) The results of the examination and tests in detail.
(b) Photographs to show damaged parts and relative movement of parts.
(c) The suppliers recommendations for design improvements to “Shock harden” the equipments.

40.6 Disposition of shock tested equipment

40.6.1 Unless otherwise specified in individual equipment specifications, shock tested equipment will be acceptable for delivery under a contract or order provided the following requirements are met:

(a) All damage resulting from shock tests is corrected. The method of correction shall be approved by the command or agency concerned.
(b) All bearings shall be replaced where examination reveals obvious but minor damage. Where major damage to bearings is discovered, appropriate redesign corrective action shall be taken.
(c) Any design improvements necessary to “Shock harden” are incorporated in the unit.
(d) The reworked unit is given the duplicate unit tests of this specification and the applicable individual equipment specification, the results are shown to be consistent with results of tests taken on other identical units.

50. ARRAINGEMENTS FOR SHOCK TESTING

50.1 Light weight and medium weight tests of accessory items (if used) Shock tests utilizing the light weight weight or medium weight shock test machines may be made at any commercial or Goverment activity having an approved test machine. The supplier shall make arrangements for such tests direct with the activity of his choice. A list of such activities having approved shock test machines in the area of the supplier’s plant may be obtained by written request to the command or agency concerned.

60. MARKING

60.1 Equipment which meets the requirements specified herein shall include the following marking on the combined-generator set installation drawing and on the detached accessory assembly drawings:

(a) Class of shockproofness.
(b) Reference to shock test reports and acceptance letter or shock design calculations and acceptance letter.
(c) Type of mounting fixutre used.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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